SmartBanana – Ripeness Classification and Quality Recommendation System For BANANANANANANANA

BMDS2133 Image Processing Assignment Documentation

Semester 202605

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Introduction

1.1 Problem Background

Bananas are among the most widely produced and consumed fruits worldwide due to their nutritional value, affordability, and year-round availability. According to the Food and Agriculture Organization (FAO), bananas represent one of the world's most important fruit crops, contributing significantly to global food security, agricultural economies, and international trade (Food and Agriculture Organization [FAO], 2023). The commercial value of bananas is determined primarily by their external appearance, ripeness stage, and surface quality, as these factors directly influence consumer purchasing behaviour, selling price, storage duration, and post-harvest losses. Consequently, accurate and consistent quality assessment has become an essential activity throughout the banana supply chain, including farms, wholesalers, supermarkets, retailers, and distribution centres.

Banana ripeness is characterised by a series of physiological and biochemical changes that occur after harvesting. During ripening, chlorophyll gradually breaks down, causing the peel colour to change from green to yellow, while brown areas develop during advanced ripening and senescence (Müller & Kräutler, 2011). Simultaneously, starch is degraded and converted into soluble sugars, contributing to changes in sweetness, texture, aroma, and overall post-harvest quality (Cordenunsi-Lysenko et al., 2019). Because banana ripeness directly determines market value and remaining shelf life, bananas sold either too early or too late may experience reduced consumer acceptance, unnecessary spoilage, and financial losses for retailers.

Despite the importance of banana quality assessment, the inspection process within many supermarkets, fruit shops, wholesalers, and packing facilities continues to rely heavily on manual visual inspection. Workers estimate ripeness by observing peel colour while simultaneously identifying visible defects such as bruises, black spots, fungal infection, mechanical damage, and surface discolouration before assigning an appropriate quality grade (Chuquimarca et al., 2024).

Problem 1: Manual banana quality assessment is subjective and inconsistent. Different inspectors may assign different ripeness stages or quality grades to the same banana because grading depends on individual experience, colour perception, and judgement. Variations in lighting conditions, viewing angles, and inspector experience further reduce grading consistency, making it difficult to maintain standardised quality control across different stores and distribution centres (Brosnan & Sun, 2004).

Problem 2: Manual inspection is labour-intensive and inefficient for large-scale operations. Commercial fruit retailers and wholesalers often process hundreds or thousands of bananas daily. Inspecting each banana individually is time-consuming and repetitive, while prolonged inspection periods increase worker fatigue and the likelihood of grading inconsistencies (Brosnan & Sun, 2004). These issues reduce operational efficiency and may result in incorrect pricing, delayed inventory rotation, unnecessary food waste, and increased post-harvest losses.

Problem 3: Existing image-based banana inspection studies commonly concentrate on individual inspection tasks, such as ripeness classification, peel-colour analysis, or spot detection, rather than integrating all quality-assessment functions into a single application (Escalante-Minakata et al., 2018; Tamatjita & Sihite, 2022; Zhang et al., 2018). Functions such as defect quantification,

quality grading, shelf-life estimation, recommendation generation, historical record management, and automated reporting are rarely combined within one system. Consequently, users may require separate processes or tools to obtain a complete banana quality assessment.

Problem 4: Many modern banana inspection systems rely heavily on deep learning models that require large labelled image datasets for training, validation, and testing. Although Convolutional Neural Networks (CNNs) and YOLO-based detectors can achieve high classification accuracy, the creation of sufficiently large and diverse fruit image datasets is time-consuming, while publicly available fruit datasets remain limited (Chuquimarca et al., 2024). These requirements may make deep learning approaches less practical for small-scale projects with limited data, development time, and computational resources.

For many practical applications involving controlled image acquisition, classical image processing provides a more transparent and computationally efficient alternative. Techniques such as image enhancement, HSV colour space conversion, thresholding, mathematical morphology, contour detection, and connected component analysis allow banana ripeness and quality to be evaluated objectively using interpretable mathematical operations (Sonka et al., 2015). Every intermediate processing result can be visualised and verified, making the entire decision-making process explainable while avoiding the computational complexity associated with deep learning.

Based on these observations, **there remains a need for an integrated, explainable, and cost-effective banana quality assessment system capable of simultaneously performing ripeness classification, defect detection, quality grading, shelf-life estimation, recommendation generation, historical data management, and automated reporting using classical image processing techniques.**

To address these problems, this project proposes **SmartBanana**, a banana ripeness classification and quality recommendation system developed using classical image processing algorithms. The proposed system acquires banana images through image upload, batch image selection, or video processing before applying image preprocessing techniques to improve image quality. HSV colour space conversion, thresholding, morphological operations, and contour detection are subsequently employed to isolate the banana from the background. The segmented fruit is analysed using colour percentage analysis to determine the ripeness stage, while threshold-based defect detection and connected component analysis objectively quantify surface damage. Based on these measurements, SmartBanana performs rule-based quality grading, estimates the remaining shelf life, generates practical recommendations, and presents the inspection results through an interactive dashboard with historical data management and automated PDF reporting. By integrating all post-harvest assessment functions into a unified, transparent, and computationally efficient workflow, the proposed system aims to provide a practical and cost-effective solution for automated banana quality assessment.

1.2 Objectives/Aims

The objectives of this project are as follows:

1. To develop an image processing-based banana ripeness assessment system capable of automatically analysing banana images and determining the ripeness stage using colour analysis techniques in the HSV colour space.

2. To evaluate banana quality by detecting surface defects, calculating the defect area percentage, and assigning quality grades through rule-based image processing techniques.

3. To generate shelf-life estimation, quality recommendations, analytical visualisations, and automated reports based on the ripeness assessment and quality grading results.

1.3 Motivation

The proposed project supports several **United Nations Sustainable Development Goals (SDGs)** by promoting sustainable agricultural practices, improving food quality management, and reducing post-harvest losses through digital innovation.

Firstly, the project contributes to **SDG 2: Zero Hunger**, which aims to end hunger, achieve food security, improve nutrition, and promote sustainable agriculture (United Nations, 2015). Post-harvest losses caused by inaccurate ripeness assessment and inefficient inventory management reduce the availability of edible food throughout the supply chain. By providing objective ripeness assessment, quality grading, and shelf-life estimation, SmartBanana assists farmers, retailers, and distributors in managing banana inventories more efficiently, thereby helping to minimise avoidable food losses and supporting a more sustainable food supply system.



Figure 1.1 SDG 2: Zero Hunger (United Nations, 2015)

Secondly, SmartBanana supports **SDG 3: Good Health and Well-being**, which seeks to ensure healthy lives and promote well-being for people of all ages (United Nations, 2015). Consumers frequently rely on visual inspection when selecting fruits, which may lead to the purchase of bananas with hidden bruises, decay, or advanced spoilage. The proposed system provides objective information regarding ripeness, visible surface defects, and remaining shelf life, enabling consumers and retailers to make better-informed decisions regarding fruit consumption and storage. This contributes to improved food quality management while encouraging safer and healthier consumption practices.



Figure 1.2 SDG 3: Good Health and Well-being (United Nations, 2015)

The proposed system also aligns with **SDG 9: Industry, Innovation and Infrastructure**, which promotes resilient infrastructure, sustainable industrialisation, and innovation (United Nations, 2015). SmartBanana demonstrates the practical application of classical image processing techniques to automate fruit quality assessment without relying on computationally intensive machine learning or deep learning models. The integration of image preprocessing, segmentation, defect detection, quality grading, dashboard visualisation, and automated reporting into a unified platform illustrates how digital technologies can modernise conventional agricultural inspection processes. Furthermore, the use of affordable imaging devices and deterministic image processing algorithms makes the proposed system accessible to small and medium-sized enterprises, encouraging wider adoption of digital technologies within the agricultural sector.



Figure 1.3 SDG 9: Industry, Innovation and Infrastructure (United Nations, 2015)

Furthermore, the project contributes directly to **SDG 12: Responsible Consumption and Production**, which emphasises the sustainable management and efficient use of natural resources while reducing food loss and waste throughout production and consumption processes (United Nations, 2015). By providing objective ripeness assessment, quality grading, shelf-life estimation, and practical recommendations for immediate sale, promotional pricing, processing, or disposal, SmartBanana assists retailers and consumers in making informed decisions regarding banana utilisation. These capabilities maximise the use of edible fruits, reduce unnecessary disposal, and encourage more responsible consumption practices that support sustainable food systems.



Figure 1.4 SDG 12: Responsible Consumption and Production (United Nations, 2015)

In addition to supporting the Sustainable Development Goals, SmartBanana is aligned with the **Malaysia MADANI** framework, particularly the principles of **Kemampanan (Sustainability)**, **Inovasi (Innovation)**, and **Kesejahteraan (Prosperity)**. Malaysia MADANI emphasises the application of digital technologies to strengthen economic resilience, improve productivity, and promote sustainable national development (Jabatan Penerangan Malaysia, 2023). By integrating image processing techniques into an automated banana quality assessment platform, SmartBanana demonstrates how technological innovation can improve traditional agricultural practices while reducing food wastage and enhancing operational efficiency. The proposed system therefore supports Malaysia's aspiration to build a digitally driven, sustainable, and innovation-oriented economy.

Overall, the motivation for developing SmartBanana extends beyond the technical implementation of image processing algorithms. The proposed system addresses practical challenges associated with manual banana quality assessment while providing economic, environmental, and societal benefits. Through objective ripeness assessment, automated defect detection, quality grading, shelf-life estimation, and intelligent recommendation generation, SmartBanana has the potential to improve decision-making throughout the banana supply chain. Furthermore, its contribution towards **SDG 2 (Zero Hunger), SDG 3 (Good Health and Well-being), SDG 9 (Industry, Innovation and Infrastructure), and SDG 12 (Responsible Consumption and Production)**, together with its alignment with the Malaysia MADANI framework, demonstrates its significance in promoting sustainable agriculture, digital innovation, responsible resource utilisation, and improved food quality management.



Figure 1.5 Malaysia Madani (Jabatan Penerangan Malaysia, 2023)

Literature Review

Scope and research context

Automated fruit inspection sits within well-established computer-vision literature because manual grading is slow, subjective, labour-intensive, and sensitive to operator inconsistency. Recent reviews of conventional imaging for fruit and vegetable quality assessment describe the workflow as a sequence of acquisition, preprocessing, segmentation, feature extraction, and decision-making, and note that RGB cameras remain the dominant sensing modality for many controlled inspection settings because they are inexpensive and sufficiently informative for external attributes such as colour, shape, and visible defects.

For bananas, that focus on visible appearance is especially appropriate. Banana peel colour changes during ripening as chlorophyll degrades, moving from green toward yellow, while advanced ripening is accompanied by surface spotting and darkening. Classical banana studies have therefore treated colour, brown area percentage, and texture as meaningful indicators of maturity and quality rather than as incidental visual cues.

This background strongly supports a non-machine-learning project architecture. In one banana study built around conventional vision features, ripening stages were described using appearance features such as CIELAB colour values, brown spot percentage, and texture. In another, HSV-based banana ripeness classification relied on colour after background removal, showing that peel chromaticity carries substantial discriminative information even before any learning-heavy pipeline is introduced.

2.1 Image segmentation for banana isolation and region extraction

Image segmentation is one of the most fundamental stages in computer vision-based fruit inspection because the accuracy of subsequent analyses, including ripeness assessment, defect detection, and quality grading, depends heavily on the successful isolation of the fruit from its background. In classical image processing, segmentation refers to the partitioning of an image into meaningful regions, enabling the system to distinguish the object of interest from surrounding pixels before quantitative measurements are performed (Sonka et al., 2015). In fruit-quality inspection, image-analysis systems commonly evaluate external characteristics such as colour, shape, size, texture, ripeness, and visible defects (Chuquimarca et al., 2024). Inaccurate segmentation can therefore reduce the reliability of colour analysis, surface-area computation, and defect quantification.

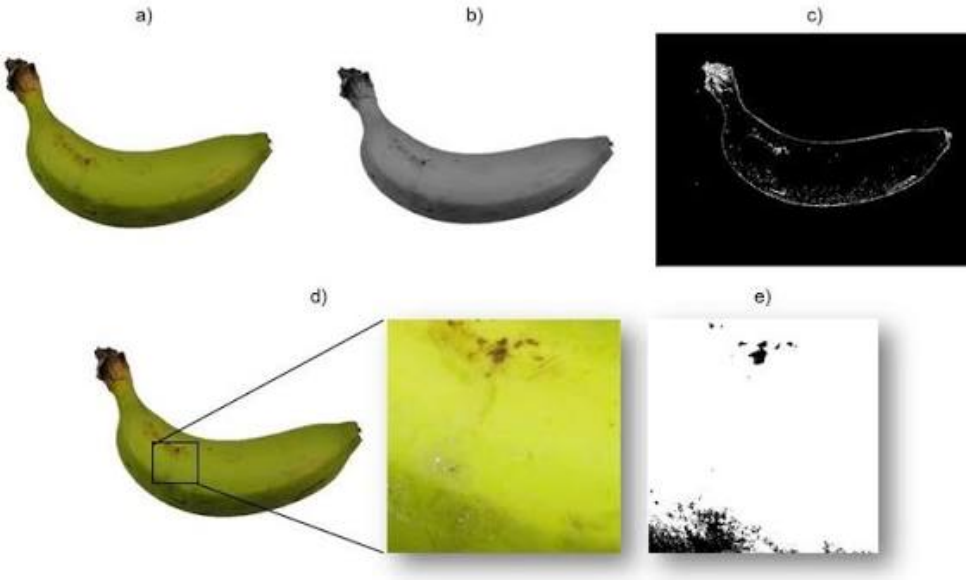


Figure 2.1 Digital image processing for banana image segmentation and region extraction: a) Original digital image, b) Transformation to grayscale, c) Contour extraction for determining fruit boundary, d) Selected square region for peel color calculation, and e) Resultant binary image detecting spot-bruise regions (Escalante-Minakata et al., 2018).

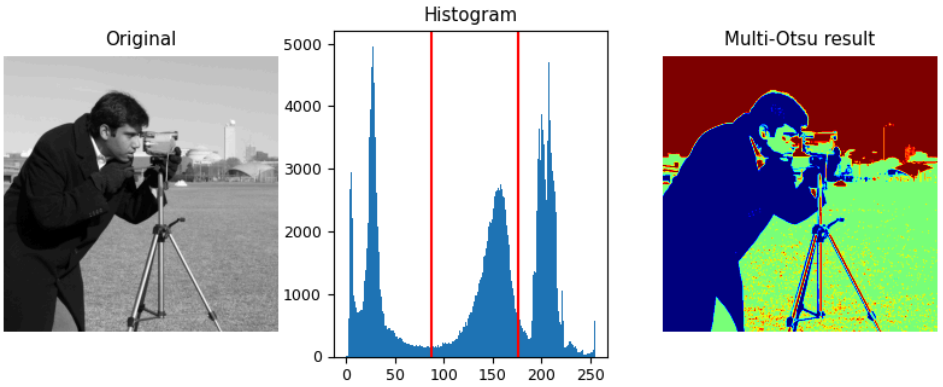


Figure 2.2 Multi-Otsu thresholding workflow: a standard greyscale image (left) is analysed via its pixel intensity histogram, where optimal thresholds are calculated (red lines, middle) to segment the scene into distinct classes (right) (Gopalakrishnan, 2023).

Threshold-based segmentation remains one of the most widely adopted approaches due to its simplicity, computational efficiency, and ease of implementation. Among these techniques, Otsu's thresholding algorithm has become the classical benchmark because it automatically determines the optimal threshold by maximising the variance between foreground and background pixel classes while minimising intra-class variance. This automatic threshold selection eliminates manual parameter tuning and produces highly reproducible segmentation results under controlled imaging conditions. Consequently, Otsu's method has been extensively employed in agricultural image processing, particularly when images are captured under uniform illumination and simple backgrounds. However, previous studies have also reported that the performance of Otsu's algorithm deteriorates when images contain uneven lighting, shadows, reflective surfaces, or complex backgrounds, as these factors distort

the grayscale histogram and reduce the separability between object and background regions (Otsu, 1979; Sonka et al., 2015).

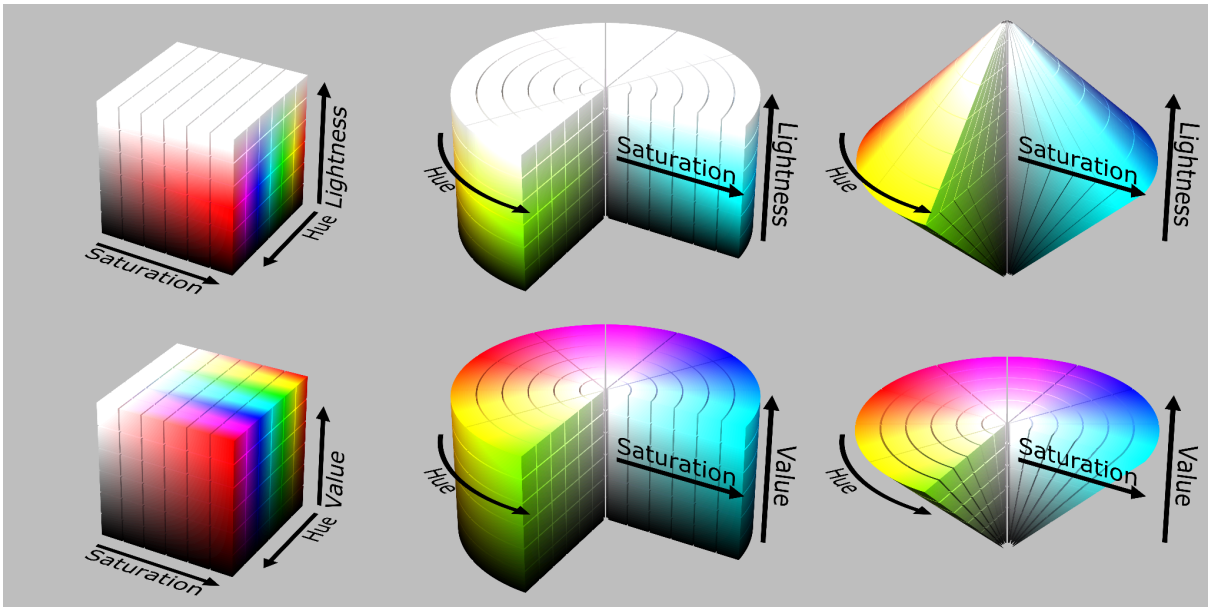


Figure 2.3 Comparison of HSL (top) and HSV (bottom) colour spaces mapped to different geometric solids (cube, cylinder, and cone) to visualise the relationship between hue, saturation, and lightness/value (SharkD & Datumizer, 2020).

To overcome the limitations associated with greyscale thresholding, colour-space-based segmentation has gained considerable attention in fruit inspection applications. Instead of operating directly on RGB images, many researchers transform images into the HSV (Hue, Saturation, Value) colour space because HSV separates chromatic information from brightness, making colour analysis less sensitive to illumination variations. This characteristic is particularly beneficial for banana ripeness assessment since banana peel colour gradually changes from green to yellow and eventually develops brown or black spots during the ripening process. Previous studies have demonstrated that HSV colour components provide strong discriminative information for identifying ripening stages while simultaneously improving the robustness of fruit segmentation under varying lighting conditions. Consequently, HSV thresholding has become one of the most frequently adopted techniques for classical banana ripeness analysis because it closely resembles human colour perception and simplifies the extraction of colour-based features.

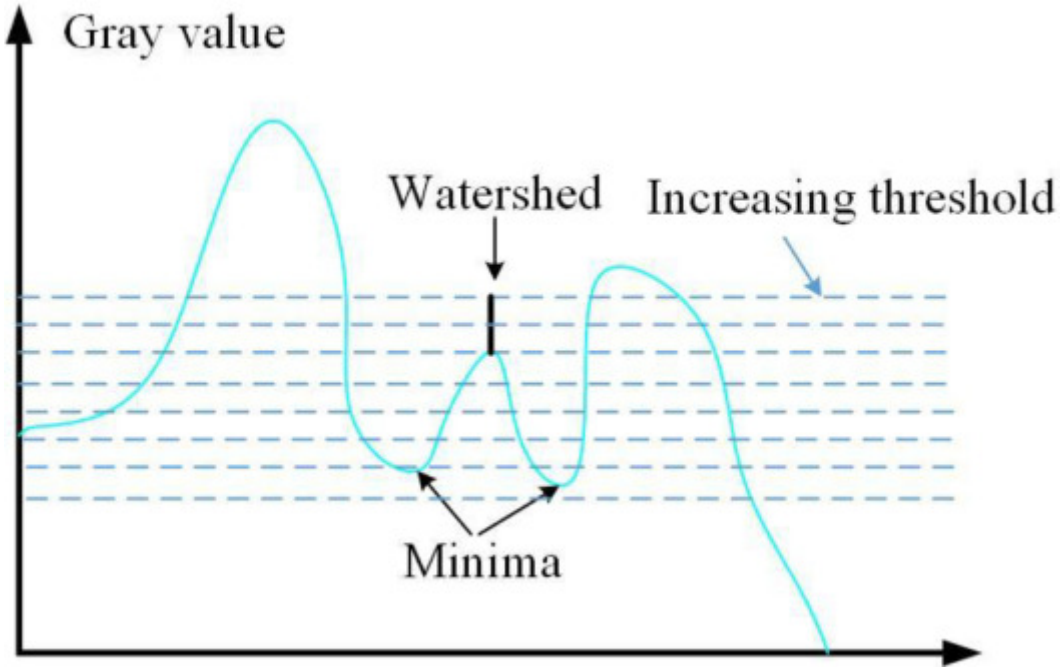


Figure 2.4 One-dimensional visualisation of the watershed segmentation concept, depicting how pixel grey values are treated as a topographical surface where water floods upward from local minima to form watershed lines at local maxima (Yuheng & Shifei, 2017).

Another widely recognised segmentation technique is watershed segmentation. Unlike thresholding approaches that rely solely on pixel intensity, watershed segmentation interprets an image as a topographical surface, where pixel intensities represent elevation values. The algorithm gradually floods the image from local minima to construct watershed boundaries that separate adjacent regions. This approach is particularly effective when multiple fruits overlap or touch one another, making it suitable for applications involving clustered agricultural products. Nevertheless, watershed segmentation is highly susceptible to over-segmentation because minor image noise often creates numerous local minima, resulting in excessive partitioning of the image. Consequently, most practical implementations employ smoothing filters, marker-controlled watershed algorithms, or morphological preprocessing before applying watershed segmentation to reduce unnecessary region fragmentation.

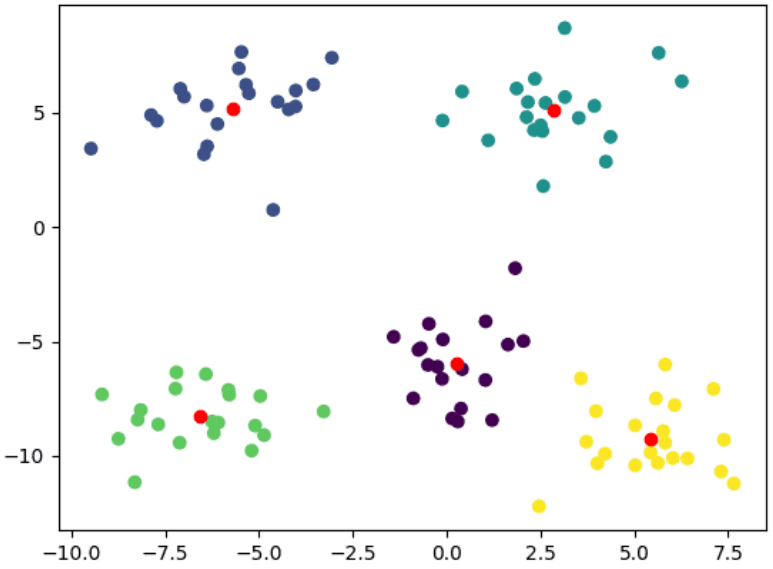


Figure 2.5 Visual representation of the K-means clustering algorithm grouping a two-dimensional dataset into five distinct, colour-coded clusters, with red markers representing the calculated cluster centroids (Cristina, 2024).

Clustering-based segmentation techniques have also become increasingly popular for fruit inspection because they are capable of handling images with complex colour distributions. K-means clustering is one of the most frequently adopted unsupervised algorithms, grouping pixels according to colour similarity within perceptually meaningful colour spaces such as CIE L*a*b*. Compared with traditional thresholding, k-means provides greater flexibility when fruit surfaces exhibit non-uniform colours or when backgrounds contain colour variations. Previous research has shown that clustering algorithms are capable of producing more homogeneous segmented regions than simple thresholding; however, they often generate fragmented regions and require additional post-processing techniques, including morphological operations or graph-based region merging, to produce inspection-ready segmentation masks. Consequently, clustering methods generally exhibit greater computational complexity than threshold-based approaches despite offering improved adaptability under challenging imaging conditions.

2.2 Contrast Enhancement For Stable Peel And Defect Analysis

Image enhancement constitutes another essential preprocessing stage in classical image processing because the effectiveness of segmentation, feature extraction, and defect analysis depends on the visual quality of the acquired images. Variations in illumination, camera exposure, surface reflections, and shadows frequently reduce image contrast, making it difficult to distinguish fruit boundaries and surface defects accurately. Consequently, numerous studies have highlighted the importance of image enhancement techniques for improving image quality before segmentation and quantitative analysis are performed.

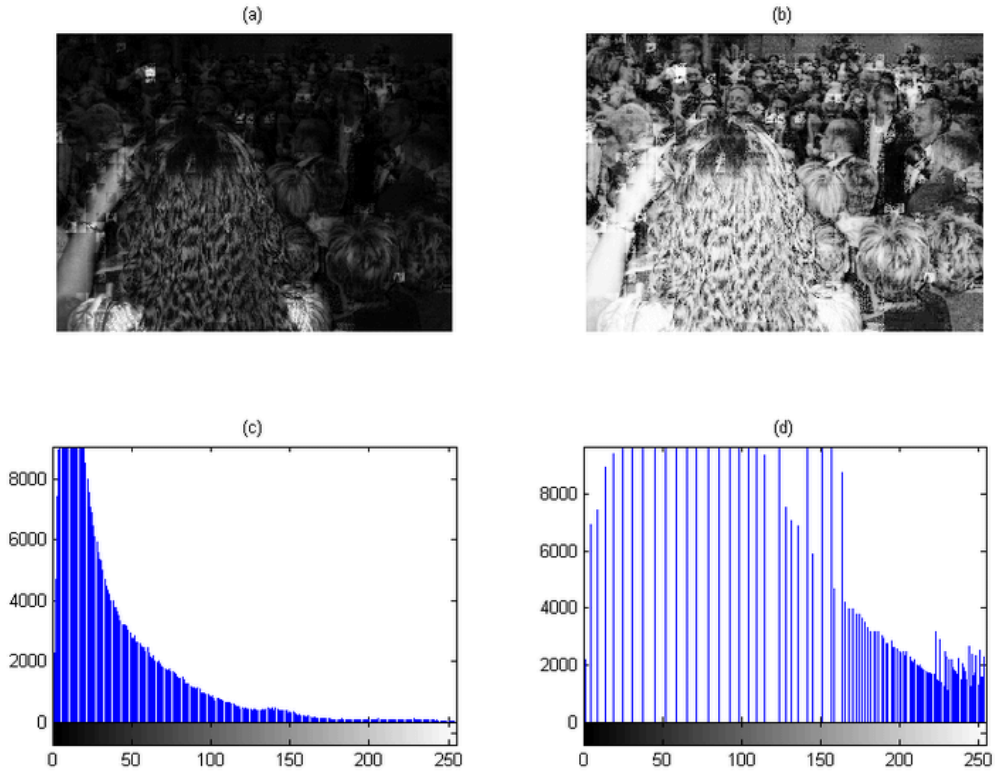


Figure 2.6 Demonstration of contrast enhancement and pixel distribution mapping: a) Original low-contrast, dark image, b) Image output after applying histogram equalisation, c) Original heavily skewed intensity histogram, and d) Broadly redistributed intensity histogram showing stretched dynamic range (Raja et al., 2015).

Histogram Equalisation (HE) is one of the earliest and most widely applied global contrast enhancement techniques (Gonzalez & Woods, 2018). The algorithm redistributes pixel intensities across the available dynamic range, thereby improving overall image contrast and making previously indistinguishable features more visible. Due to its computational simplicity and rapid execution, histogram equalisation has been extensively utilised in image processing applications involving medical imaging, industrial inspection, and agricultural quality assessment. Nevertheless, because the transformation is applied globally across the entire image, local contrast differences are often not sufficiently enhanced. Furthermore, excessive histogram stretching may amplify image noise and distort brightness, particularly when images contain large homogeneous regions or significant illumination variation.

Adaptive Histogram Equalisation (AHE) was subsequently introduced to address the shortcomings of global histogram equalisation by performing local contrast enhancement within small neighbourhoods rather than across the entire image (Pizer et al., 1987). This localised enhancement enables subtle features, including fruit edges, small bruises, and texture variations, to become significantly more distinguishable. Despite these advantages, previous studies have reported that AHE frequently amplifies random image noise within relatively uniform regions because local histograms are independently enhanced throughout the image. In addition, the computational cost of AHE is considerably higher than conventional histogram equalisation, making it less suitable for real-time processing or large image datasets.

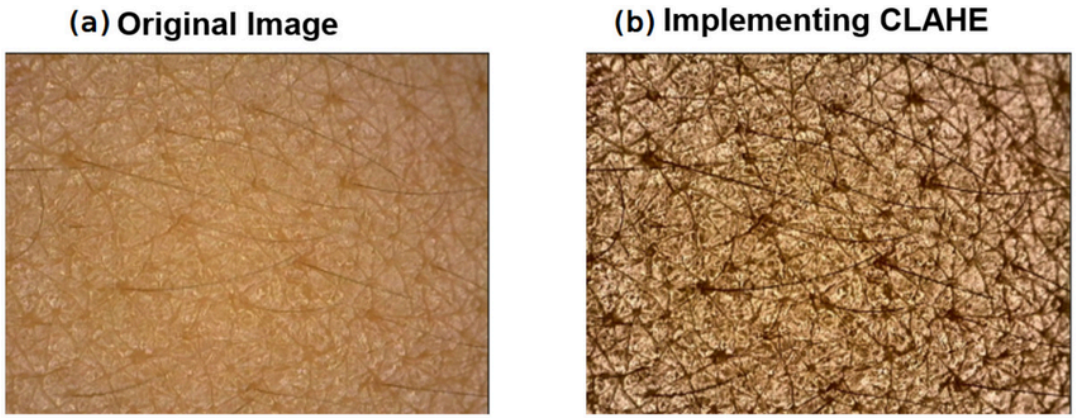


Figure 2.7 Comparison of a local texture image: a) Original low-contrast image, and b) Enhanced texture, fine-line, and feature visibility after applying Contrast Limited Adaptive Histogram Equalisation (CLAHE) (Saiwaeo et al., 2023).

To overcome these limitations, Contrast Limited Adaptive Histogram Equalisation (CLAHE) was developed as an improved version of AHE. Instead of allowing unrestricted histogram amplification, CLAHE introduces a clip limit that prevents excessive enhancement within homogeneous image regions. Consequently, CLAHE significantly improves local contrast while suppressing unnecessary noise amplification. Numerous agricultural vision studies have demonstrated that CLAHE effectively enhances subtle fruit surface features, particularly under non-uniform illumination, making it one of the most widely adopted enhancement techniques in modern fruit inspection systems. Although CLAHE requires greater computational resources than conventional histogram equalisation, its superior robustness under challenging lighting conditions generally outweighs the additional processing cost.

Retinex-based enhancement represents another important category of illumination correction techniques. Unlike histogram-based approaches that primarily redistribute pixel intensities, Retinex algorithms estimate the illumination component of an image and separate it from surface reflectance. This process produces images with improved colour constancy and enhanced visibility under uneven lighting conditions. Several studies have demonstrated that Retinex-based algorithms effectively preserve colour fidelity while simultaneously improving local contrast, making them suitable for outdoor agricultural imaging where illumination changes continuously. However, Retinex algorithms involve substantially greater mathematical complexity and require careful parameter optimisation, making them less practical for undergraduate projects that prioritise simplicity, interpretability, and computational efficiency.

2.3 Mathematical Morphology and Connected-component Analysis For Defect Quantification

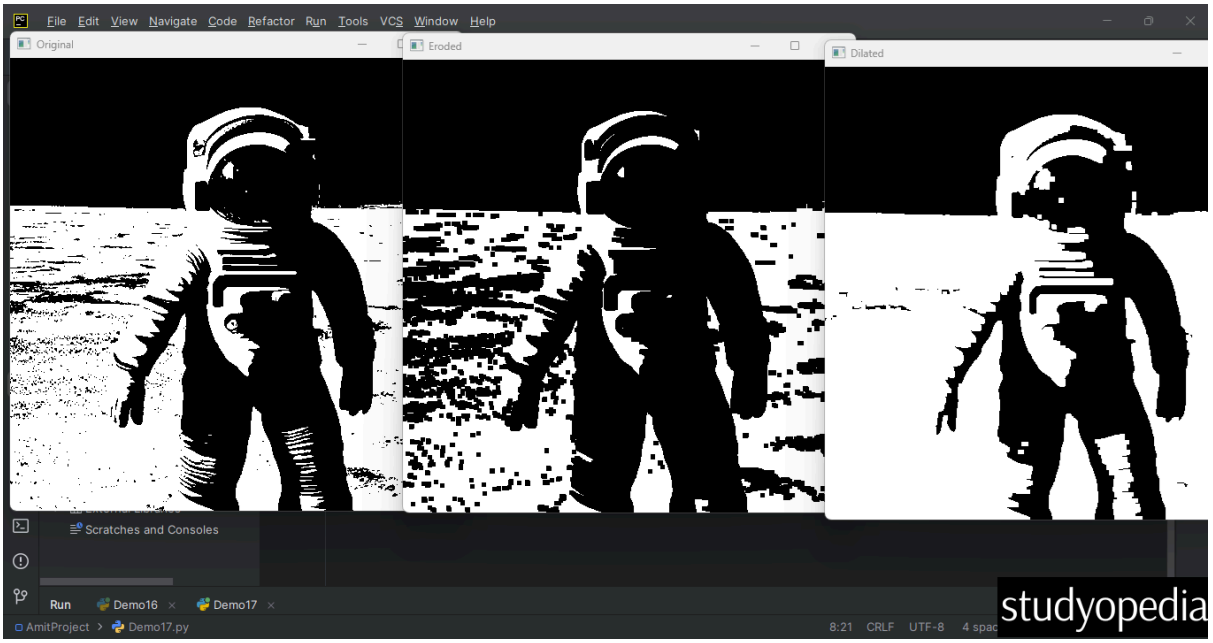


Figure 2.8 Basic morphological operations on a binary mask: showing the original thresholded silhouette (left), the eroded result with reduced foreground boundaries (middle), and the dilated result with expanded foreground boundaries (right) (Studyopedia, 2025a).

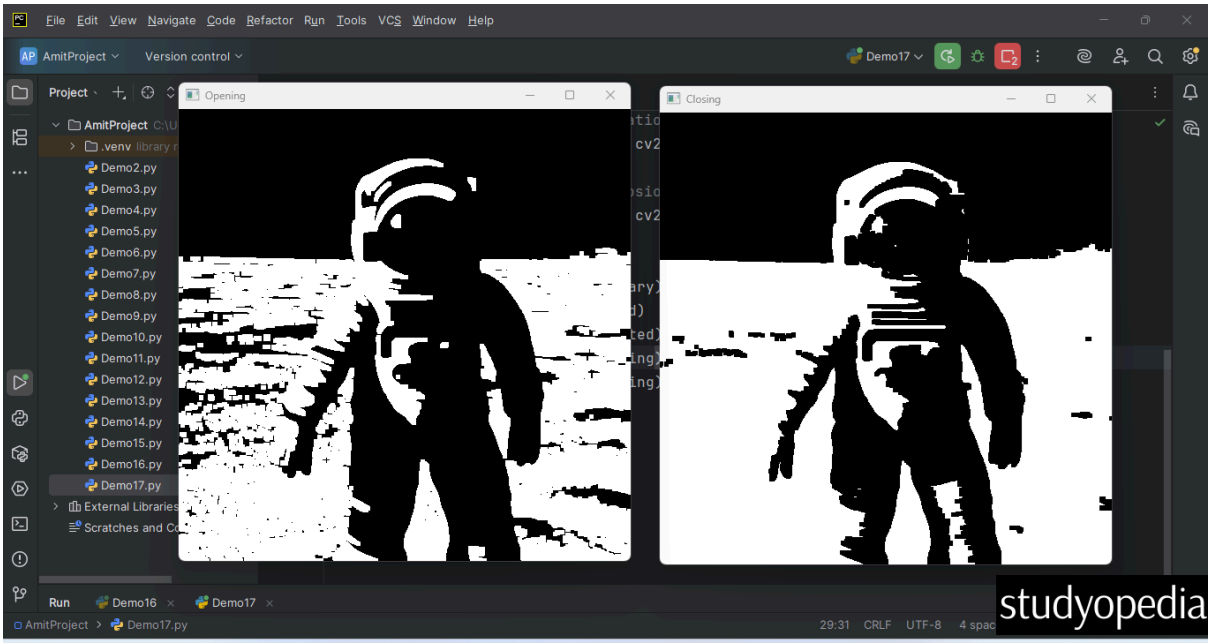


Figure 2.9 Compound morphological operations: showing the opening operation which effectively removes isolated background noise (left), and the closing operation which successfully fills internal gaps and holes within the target silhouette (right) (Studyopedia, 2025b).

Mathematical morphology is a fundamental image processing technique that analyses and modifies image structures according to their geometric properties (Serra, 1982). Following image segmentation, binary masks frequently contain imperfections such as isolated noise pixels, small holes, irregular boundaries, and fragmented object regions. These artefacts reduce the accuracy of subsequent measurements, including fruit-area estimation, contour extraction, colour analysis, and defect quantification. Consequently, morphological operations are commonly applied as a post-processing stage to refine segmentation masks before feature extraction and object analysis are performed.

Morphological processing is based on the application of predefined structuring elements that probe the geometric properties of binary images. Four fundamental operations form the basis of morphological analysis: erosion, dilation, opening, and closing. Erosion removes boundary pixels, thereby eliminating small isolated objects and reducing object size. Dilation performs the opposite operation by expanding object boundaries and filling narrow gaps between neighbouring pixels. Opening, which consists of erosion followed by dilation, effectively removes small background noise while preserving the overall shape of larger objects. Conversely, closing, which applies dilation followed by erosion, fills small holes and repairs discontinuities within segmented regions. These complementary operations significantly improve the geometric integrity of binary masks, allowing subsequent measurements to represent the true physical characteristics of the fruit surface.

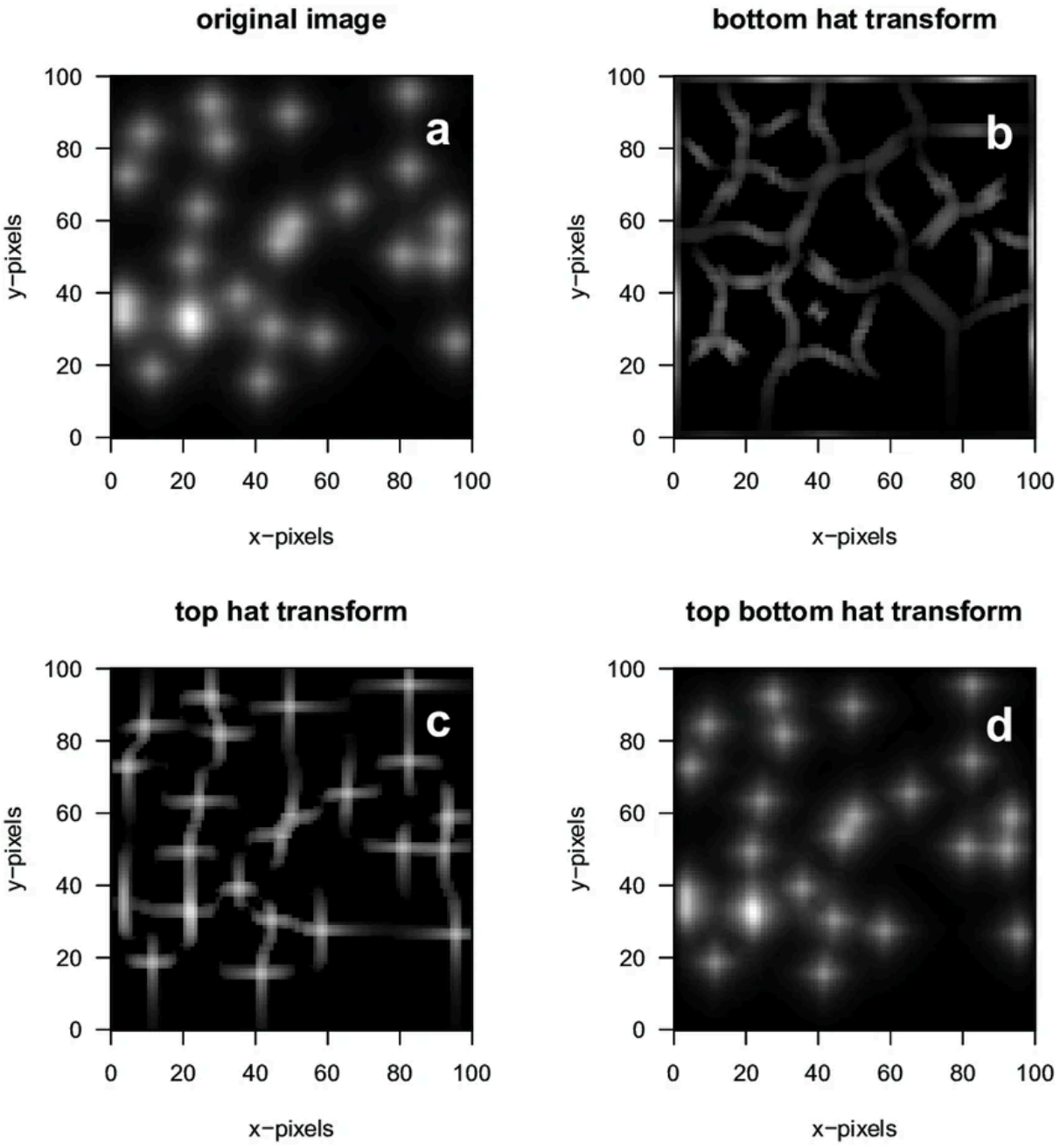


Figure 2.10 Advanced morphological feature isolation using a synthetic greyscale test image: a) Original simulated image, b) Bottom-hat transform isolating dark interstitial valleys, c) Top-hat transform isolating localized bright peaks, and d) Combined top-bottom hat transformation for local feature enhancement (Wagner et al., 2019).

Beyond the four basic operations, more advanced morphological transformations further enhance defect detection. The top-hat transformation extracts bright objects that are smaller than the structuring element, whereas the bottom-hat transformation highlights dark objects against brighter backgrounds (Gonzalez & Woods, 2018; Serra, 1982). Since overripe bananas commonly develop brown bruises, black spots, and localised dark patches, bottom-hat morphology is particularly valuable because it increases the visibility of these darker surface defects without significantly affecting healthy peel regions. Consequently, defect regions become easier to isolate through thresholding and subsequent quantitative analysis.

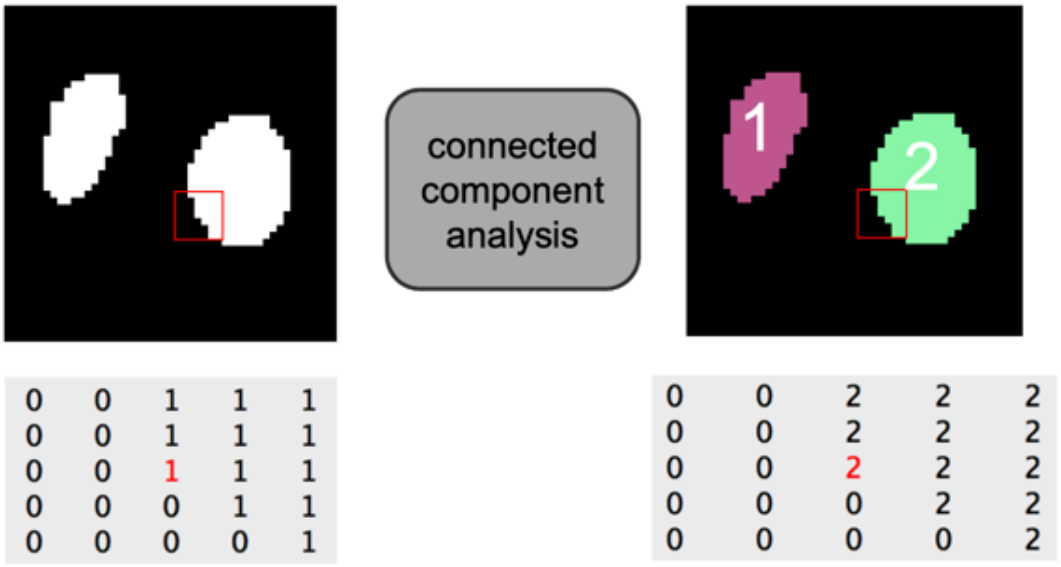


Figure 2.11 The mechanism of Connected Component Analysis (CCA): transforming a raw binary mask (left) into a multi-class label image (right) where distinct pixel groups are assigned unique integer identifiers based on spatial connectivity (EMBL-community, 2026).

Once morphological processing has produced a clean binary mask, Connected Component Analysis (CCA) provides the transition from image segmentation to quantitative feature extraction. Connected Component Analysis assigns a unique label to every connected region within a binary image, allowing individual objects or defect regions to be analysed independently (He et al., 2009). Rather than treating all detected defects as a single region, CCA separates each bruise or black spot into an individual connected component.

2.4 Critical Comparison of Existing Approaches

Numerous image processing techniques have been proposed for fruit inspection. However, no single algorithm consistently achieves optimal performance across all imaging environments. The effectiveness of a particular technique depends largely on image acquisition conditions, illumination consistency, background complexity, fruit overlap, and the intended inspection objective. Consequently, previous studies have emphasised selecting algorithms according to the characteristics of the inspection environment rather than adopting the most sophisticated method available.

Threshold-based segmentation methods remain among the most popular approaches because they are computationally efficient, straightforward to implement, and highly interpretable. Otsu's thresholding automatically determines an optimal segmentation threshold and performs exceptionally well when images are captured against uniform backgrounds under stable lighting conditions. Nevertheless, the algorithm becomes increasingly unreliable when images exhibit non-uniform illumination, shadows, reflections, or overlapping intensity distributions. Under such circumstances, segmentation errors may significantly reduce the accuracy of subsequent colour analysis and defect measurement.

Consequently, Otsu's algorithm is generally recommended only after suitable colour-space conversion or image enhancement has been applied.

Watershed segmentation offers substantially greater flexibility by modelling images as topographical surfaces and identifying boundaries through simulated flooding processes. This characteristic enables watershed algorithms to separate touching or overlapping fruits that cannot easily be isolated using conventional thresholding. Despite these advantages, watershed segmentation frequently suffers from over-segmentation because even minor image noise generates additional local minima, producing an excessive number of segmented regions. Practical implementations therefore require additional preprocessing, including Gaussian smoothing, marker generation, or morphological filtering, increasing computational complexity and implementation difficulty.

Clustering algorithms, particularly k-means clustering, represent another widely investigated segmentation strategy. Instead of relying on manually selected thresholds, k-means groups pixels according to colour similarity, allowing the algorithm to accommodate more complex colour distributions and varying backgrounds. Studies have demonstrated that clustering techniques generally outperform fixed thresholding when fruit surfaces exhibit heterogeneous colour characteristics. However, clustering algorithms often produce fragmented regions, ignore spatial continuity between neighbouring pixels, and require subsequent region merging or morphological refinement before meaningful quantitative measurements can be performed. As a result, clustering approaches typically require greater computational resources than threshold-based segmentation while providing only moderate improvements under controlled imaging conditions.

Image enhancement techniques exhibit similar trade-offs. Global Histogram Equalisation provides rapid improvement of overall image contrast but frequently fails to enhance local details associated with bruises, texture variations, and subtle colour transitions. Adaptive Histogram Equalisation improves local contrast considerably but often amplifies image noise within homogeneous regions. CLAHE successfully addresses these shortcomings by limiting excessive contrast amplification while preserving local image detail. Consequently, CLAHE has become the preferred enhancement technique in numerous agricultural vision applications involving variable illumination and subtle surface defects.

The literature therefore indicates that algorithm selection should prioritise simplicity, robustness, and interpretability over algorithmic complexity whenever images are acquired under controlled laboratory conditions. Sophisticated segmentation algorithms provide measurable advantages primarily when imaging environments involve overlapping fruits, highly variable illumination, or complex backgrounds. In contrast, classical pipelines combining CLAHE, HSV thresholding, morphological refinement, contour analysis, and connected component analysis consistently provide reliable performance for single-fruit inspection while remaining computationally efficient and easy to interpret.

Table 2.1 Comparison of Image Segmentation Techniques

Algorithm / System	Example of Previous Work	Advantages	Limitations	Remarks

	(insert as citation)			
Algorithm 1: Otsu Thresholding	Otsu (1979)	Fast, automatic, computationally efficient, reproducible, simple implementation.	Sensitive to uneven illumination and non-bimodal histograms.	Highly suitable because banana images are captured under controlled conditions.
Algorithm 2: HSV Colour-Based Analysis	Tamatjita and Sihite (2022)	Separates chromatic information from brightness and directly analyses banana peel colour.	HSV ranges may require calibration for different banana varieties, cameras, and lighting conditions.	Suitable because banana ripeness is strongly associated with peel-colour changes.
Algorithm 3: Watershed Segmentation	Vincent and Soille (1991)	Excellent for separating touching objects.	High risk of over-segmentation; requires extensive preprocessing.	Suitable only when multiple bananas overlap.
Algorithm 4: K-Means Clustering	Cristina (2024)	Handles complex colour distributions and heterogeneous surfaces.	Computationally intensive; frequently requires post-processing.	Suitable for uncontrolled environments but unnecessary for single-fruit analysis.

Table 2.2 Comparison of Contrast Enhancement Techniques

Algorithm / System	**Example of Previous Work *(insert as citation)***	**Advantages**	**Limitations**	**Remarks**
Algorithm 1: Histogram Equalisation (HE)	Pizer et al (1987)	Fast, simple implementation , improves global image contrast.	Poor local enhancement; may amplify brightness inconsistently.	Suitable only under uniform lighting.
Algorithm 2: Adaptive Histogram Equalisation (AHE)	Pizer et al. (1987)	Enhances local image details effectively.	Amplifies image noise; higher computational complexity.	Useful but not recommended for final implementation .
Algorithm 3: CLAHE	Zuiderveld (1994)	Enhances local contrast while limiting noise amplification; robust under varying illumination.	Requires parameter tuning; slightly higher computational cost.	Selected for the proposed system because it provides the best balance between robustness and computational efficiency.
Algorithm 4: Retinex Enhancement	Jobson et al. (1997) Rahman et al. (2004)	Excellent illumination correction and colour constancy.	Computationall y complex and difficult to optimise.	Better suited to advanced research rather than undergraduate coursework.

Table 2.3 Comparison of Defect Detection Algorithms

Algorithm / System	**Example of Previous Work *(insert as citation)***	**Advantages**	**Limitations**	**Remarks**
Algorithm 1: Threshold-Based Defect Detection	Escalante-Minakata et al. (2018)	Simple implementation, fast computation, suitable for obvious defects.	Sensitive to illumination changes, may miss subtle bruises.	Suitable for detecting dark brown and black banana spots after colour-based segmentation.
Algorithm 2: Morphological Processing	Serra (1982)	Removes segmentation noise, repairs fragmented objects, and improves defect localisation.	Performance depends on structuring element selection, may remove very small defects if parameters are unsuitable.	Essential post-processing stage after thresholding to improve defect mask quality.
Algorithm 3: Connected Component Analysis (CCA)	He et al. (2009)	Counts individual defects, measures defect area accurately, supports quantitative quality grading.	Requires accurate binary segmentation beforehand, cannot distinguish defect types without additional analysis.	Selected because SmartBanana grades bananas according to total damaged surface percentage rather than merely detecting defects.

2.5 Research Gap

Despite considerable progress in automated fruit inspection, several limitations remain evident within the existing literature. First, the majority of recent studies concentrate primarily on ripeness classification while paying relatively little attention to simultaneous defect detection and overall quality grading. Although colour-based maturity assessment has been extensively investigated, quantitative measurement of bruises, black spots, and surface damage is often omitted or treated as a secondary objective. Consequently, many proposed systems are capable of determining ripeness stages but cannot comprehensively evaluate the commercial quality of individual fruits.

Second, most published systems employ deep learning techniques that require extensive annotated datasets and computationally intensive model training. While these methods achieve remarkable classification accuracy, they are generally unsuitable for educational environments that emphasise understanding of classical image processing algorithms. Furthermore, the computational requirements of deep learning models may limit their deployment on low-cost hardware and reduce system interpretability. In contrast, relatively few studies investigate fully classical computer vision pipelines capable of producing accurate, explainable, and computationally efficient fruit inspection systems.

Third, many existing studies perform evaluation using only individual images collected under laboratory conditions. Features such as batch image analysis, automated reporting, dashboard visualisation, shelf-life estimation, and recommendation support remain uncommon despite their practical importance for agricultural businesses and fruit retailers. Consequently, many published solutions remain research prototypes rather than complete end-user applications capable of supporting operational decision-making.

Finally, explainability remains a significant challenge in modern computer vision systems. Deep learning models frequently provide only final classification outputs without revealing the underlying visual evidence supporting each decision. Such limited interpretability may reduce user confidence and complicate model validation, particularly in educational contexts where algorithm understanding forms a major learning objective. Classical image processing techniques address this limitation by allowing every intermediate processing stage to be visualised and quantitatively analysed.

Therefore, the proposed SmartBanana system addresses these research gaps by integrating ripeness assessment, defect quantification, quality grading, shelf-life prediction, recommendation support, and comprehensive reporting within a single classical image processing framework. Unlike many previous studies, every decision generated by the proposed system can be directly explained through measurable image features, making the system both scientifically interpretable and practically applicable.

Methodology

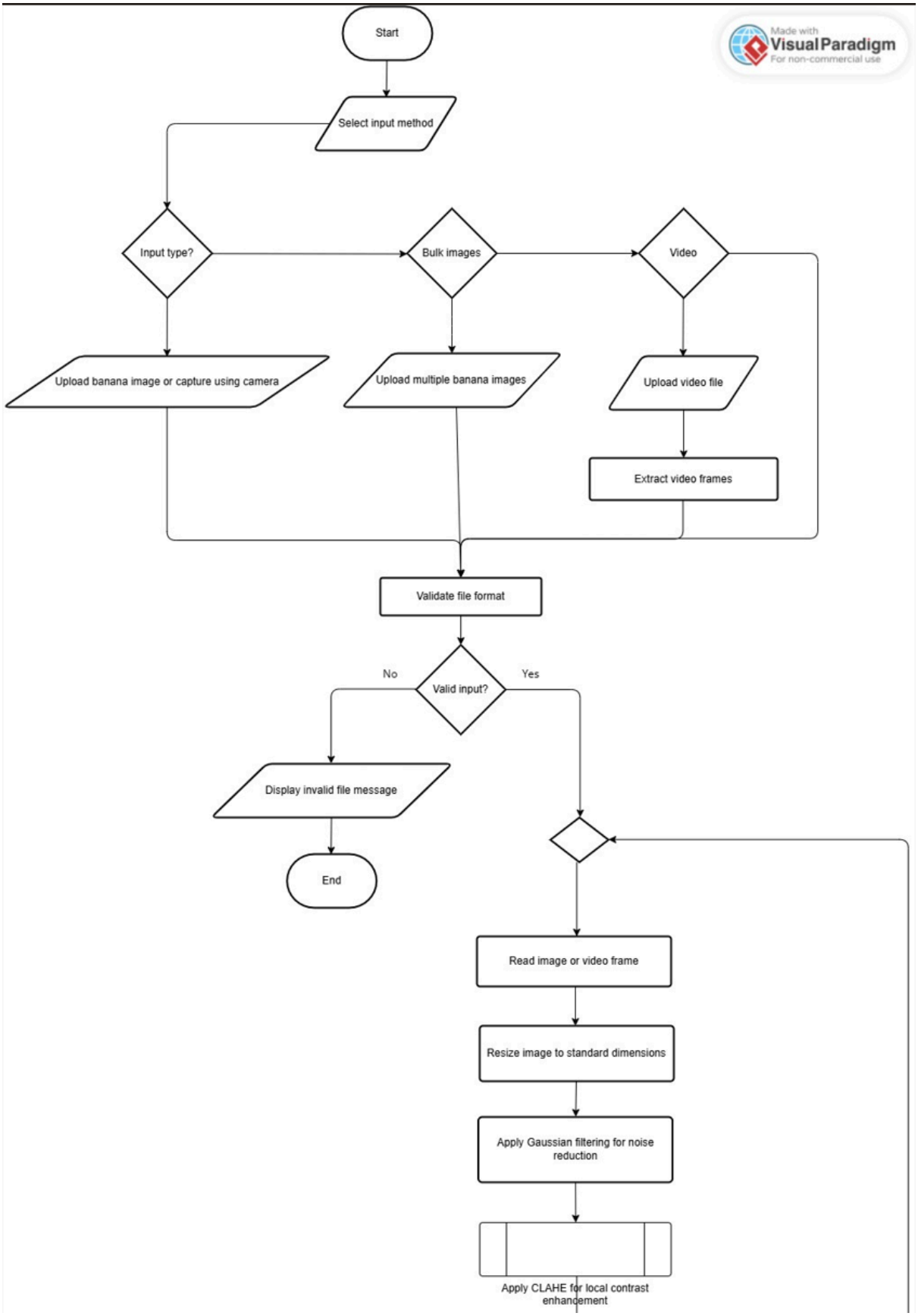
3.1 System flowchart

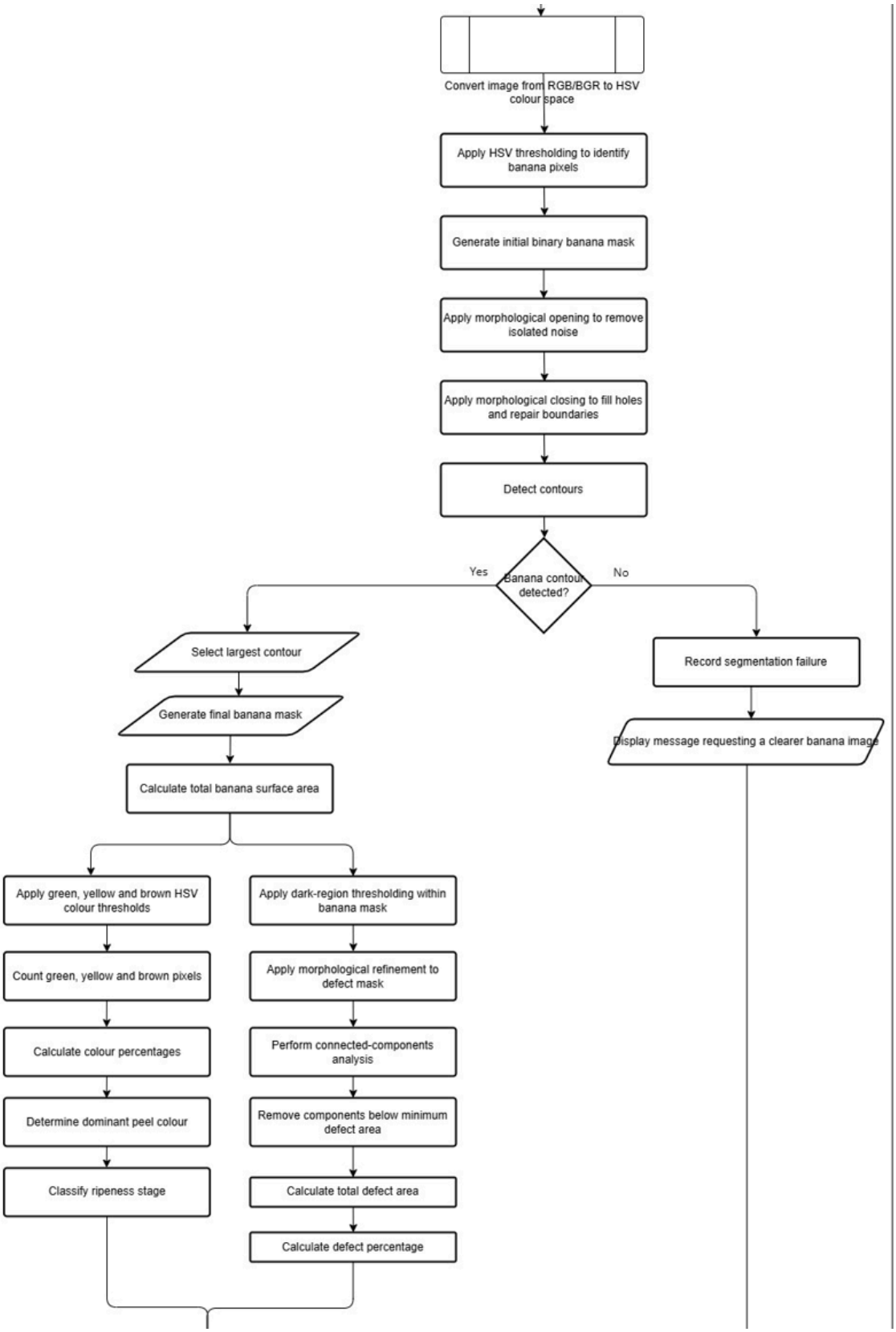
The proposed SmartBanana system follows a structured image processing pipeline consisting of image acquisition, preprocessing, banana segmentation, ripeness analysis, defect detection, quality grading, shelf-life prediction, recommendation generation, and report management. Rather than relying on deep learning models, the proposed methodology employs deterministic image processing algorithms that provide transparent and explainable decision-making while maintaining low computational complexity. Each processing stage generates intermediate outputs that can be visualised and verified, allowing users to understand how the final quality assessment is produced. The overall workflow is illustrated in Figure 3.1.

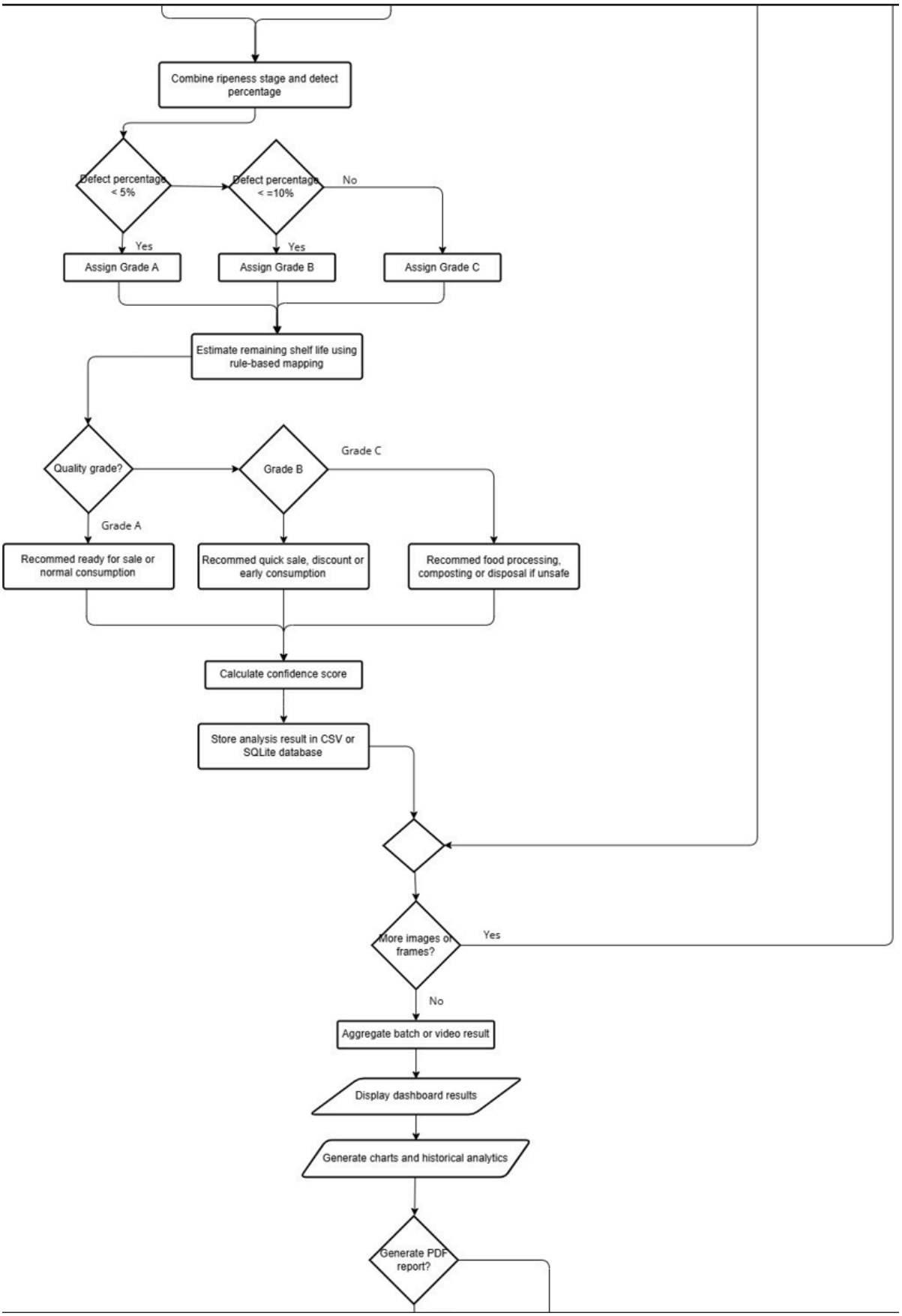
https://drive.google.com/file/d/1WV2_a86RMfLL39SvkFTrzhnMuGygQYTl/view?usp=sharing

SmartBanana - System Flowchart.jpg

https://online.visual-paradigm.com/share.jsp?id=343353373239362d3230


Start
Select input method
Input type?
Bulk images
Video
Upload banana image or capture using camera
Upload multiple banana images
Upload video file
Extract video frames
Validate file format
Valid input?
No
Yes
Display invalid file message
End
Read image or video frame
Resize image to standard dimensions
Apply Gaussian filtering for noise reduction
Apply CLAHE for local contrast enhancement
Made with VisualParadigm
For non-commercial use


Convert image from RGB/BGR to HSV colour space
Apply HSV thresholding to identify banana pixels
Generate initial binary banana mask
Apply morphological opening to remove isolated noise
Apply morphological closing to fill holes and repair boundaries
Detect contours
Banana contour detected?
Yes
No
Select largest contour
Generate final banana mask
Calculate total banana surface area
Record segmentation failure
Display message requesting a clearer banana image
Apply green, yellow and brown HSV colour thresholds
Count green, yellow and brown pixels
Calculate colour percentages
Determine dominant peel colour
Classify ripeness stage
Apply dark-region thresholding within banana mask
Apply morphological refinement to defect mask
Perform connected-components analysis
Remove components below minimum defect area
Calculate total defect area
Calculate defect percentage


Combine ripeness stage and detect percentage
Defect percentage < 5%
Defect percentage < =10%
No
Yes
Yes
Assign Grade A
Assign Grade B
Assign Grade C
Estimate remaining shelf life using rule-based mapping
Quality grade?
Grade B
Grade C
Grade A
Recommed ready for sale or normal consumption
Recommed quick sale, discount or early consumption
Recommed food processing, composting or disposal if unsafe
Calculate confidence score
Store analysis result in CSV or SQLite database
More images or frames?
Yes
No
Aggregate batch or video result
Display dashboard results
Generate charts and historical analytics
Generate PDF report?

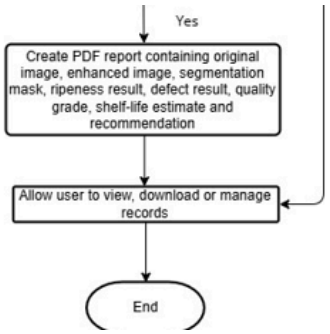


Figure 3.1: SmartBanana System Architecture and Image Processing Flowchart.

Figure 3.1 illustrates the complete processing pipeline implemented in the SmartBanana system. The workflow begins when the user uploads an image from the local device or captures a new banana image using the system camera. The acquired image is then forwarded to the image preprocessing stage, where resizing, Gaussian filtering, and Contrast Limited Adaptive Histogram Equalisation (CLAHE) are applied. These operations improve image quality by standardising image dimensions, suppressing sensor noise, and enhancing local contrast, thereby producing a more reliable input for subsequent image analysis.

Following preprocessing, the enhanced RGB image is converted into the HSV colour space. Unlike RGB, HSV separates colour information from brightness, making the segmentation process less sensitive to illumination changes. HSV thresholding is then applied to distinguish banana pixels from the background based on predefined colour ranges corresponding to banana peel colours.

The resulting binary mask may still contain isolated noise, small holes, and fragmented object boundaries. Therefore, morphological opening and closing operations are applied to remove small artefacts while repairing discontinuities within the segmented banana. After morphological refinement, contour detection is performed to identify all connected objects in the binary image. Since only one banana is analysed at a time, the largest contour is selected as the banana object while smaller background objects are discarded. This process produces an accurate banana mask that serves as the basis for all subsequent analyses.

Once the banana has been successfully segmented, the system simultaneously performs ripeness analysis and defect detection. During ripeness analysis, the segmented banana is analysed using HSV colour thresholds corresponding to green, yellow, and brown peel colours. The percentage of pixels belonging to each colour category is calculated to objectively determine the maturity stage of the banana. Based on predefined decision rules, the system classifies the banana into one of several ripeness categories, including unripe, partially ripe, ripe, or overripe.

At the same time, defect detection is performed by identifying dark regions that correspond to bruises, black spots, fungal infection, or surface deterioration. Binary thresholding is applied to isolate potential defects, followed by morphological refinement to eliminate false detections. Connected Component Analysis is subsequently employed to identify each defect as an independent connected

region. The total defect area is calculated as a percentage of the entire banana surface, providing an objective numerical indicator of fruit quality.

The ripeness classification and defect percentage are subsequently integrated within a rule-based quality grading module. Rather than relying on subjective human judgement, the grading rules combine colour information and surface damage measurements to classify bananas into predefined quality categories. This transparent grading mechanism ensures that every decision can be directly explained through measurable image features.

Following quality grading, the system estimates the remaining shelf life according to the detected ripeness stage and quality condition. A recommendation engine then generates practical suggestions for storage, consumption, selling, or processing based on the predicted shelf life. For example, bananas that are fully ripe may be recommended for immediate consumption, whereas overripe bananas may be suggested for baking or smoothie preparation.

Finally, all inspection results are presented through the dashboard module, where users can review the original image, segmentation results, ripeness classification, defect percentage, quality grade, shelf-life estimation, and recommendations. The system also supports batch image processing, video analysis, PDF report generation, and historical record management, enabling users to analyse multiple banana images efficiently while maintaining complete inspection records for future reference.

Overall, the proposed methodology integrates image enhancement, segmentation, feature extraction, morphology, object analysis, and decision-making into a unified image processing pipeline. Compared with manual inspection, the proposed workflow provides more objective, consistent, and explainable quality assessment while satisfying the assignment requirement of applying classical image processing techniques throughout the system implementation.

3.2 Description of dataset

The SmartBanana system uses a combination of publicly available banana images and self-collected images to support algorithm calibration, functional testing, and experimental evaluation. Although the proposed system is based primarily on classical image processing rather than supervised machine learning, a varied image collection remains necessary to determine appropriate HSV threshold ranges, evaluate segmentation accuracy, test defect detection, and assess the reliability of the final ripeness and quality grading rules.

The selected public image source is the **FreshCheck Fruit Ripeness Classification Dataset**, which contains banana and tomato images organised according to different ripeness and quality conditions. Only the banana subset is used in the SmartBanana project because the system is specifically designed for banana ripeness and surface quality assessment. The dataset provides images representing different peel colour conditions, including green, partially yellow, fully yellow, and heavily spotted or overripe bananas. These visual differences are suitable for evaluating the proposed colour-based ripeness analysis process.

Dataset source: FreshCheck Fruit Ripeness Classification Dataset
Dataset platform: Kaggle
Dataset URL: https://www.kaggle.com/datasets/shahriar26s/banana-ripeness-classification-dataset
Image formats: JPG, JPEG, and PNG

Banana images: Approximately 13,478 images
Ripeness classes: Four classes
Planned data division: 80% testing and calibration, 10% validation, and 10% final evaluation

The dataset is not used to train an artificial intelligence model. Instead, representative images from each ripeness class are examined to determine suitable colour thresholds and to test whether the classical image processing pipeline can distinguish the major banana ripeness stages. This approach remains consistent with the requirement to apply image enhancement, filtering, segmentation, feature extraction, morphology, and object analysis techniques.

3.2.1 Dataset Categories

The banana images are grouped into four general ripeness categories according to their visible peel characteristics. The categories used by SmartBanana are shown in Table 3.1.

Table 3.1: Banana Ripeness Categories

Category	Visible Peel Characteristics	SmartBanana Output
Unripe	Peel is predominantly green with little or no yellow colour	Unripe
Ripe	Peel is predominantly yellow with little or no brown spotting	Ripe
Overripe	Peel is yellow with noticeable brown spots indicating advanced ripening	Overripe
Spoiled	Peel contains extensive black or dark brown regions, decay, or severe deterioration	Spoiled

The banana images are organised into four categories according to the FreshCheck dataset: unripe, ripe, overripe, and spoiled. These categories represent different stages of banana maturity and deterioration based on external peel appearance. The SmartBanana system uses these labelled images to evaluate the effectiveness of the proposed image processing algorithms. During processing, the colour distribution of the banana peel and the percentage of surface defects are analysed to determine which category most closely matches the input image. The spoiled category differs from the overripe category because it represents bananas with severe surface deterioration, extensive blackening, fungal growth, or decay that significantly reduces their quality and remaining shelf life.

3.2.2 Self-Collected Images

In addition to the public dataset, self-collected banana images are included to evaluate the prototype under realistic conditions. These images are captured using ordinary mobile phone cameras and

computer webcams. Bananas at different ripeness stages are photographed under several lighting conditions, image distances, camera angles, and backgrounds.

The self-collected images are used for the following purposes:

1. Testing whether the system can process images captured outside the original dataset environment.
2. Evaluating the effect of illumination variation on HSV thresholding.
3. Testing the effectiveness of Gaussian filtering and CLAHE enhancement.
4. Assessing whether bruises, brown spots, and black surface regions can be detected.
5. Evaluating single-image, batch-image, and video-frame processing.
6. Identifying cases in which the segmentation or grading rules require further parameter adjustment.

To reduce unnecessary variability, the recommended image acquisition environment uses a plain contrasting background, sufficient lighting, minimal shadow, and a single banana positioned near the centre of the image. The entire banana should be visible and should not be heavily covered by other objects. Images that are excessively blurred, underexposed, overexposed, or incomplete may be rejected by the input validation module.

3.2.3 Image Acquisition Requirements

For reliable processing, each image should satisfy the following acquisition requirements:

- The input must use the JPG, JPEG, or PNG format.
- At least one clearly visible banana must appear in the image.
- The fruit should occupy a substantial portion of the image.
- The entire banana surface should be visible whenever possible.
- The image should contain sufficient lighting to distinguish green, yellow, and brown peel regions.
- Severe motion blur should be avoided.
- A background that contrasts with the banana peel is recommended.
- The image should not contain strong yellow objects that may be incorrectly included within the segmentation mask.

All accepted images are converted into a standard internal format before processing. Colour images are read as three-channel arrays, resized to a fixed maximum dimension, and stored temporarily as NumPy arrays for OpenCV processing. Resizing standardises the computational workload while preserving the original aspect ratio.

3.2.4 Data Structure

Each image analysis produces a structured record containing the input details, extracted image features, and final assessment results. The proposed data structure is shown in Table 3.2.

Table 3.2: SmartBanana Data Dictionary

Field Name	Data Type	Description
record_id	Integer/String	Unique identifier assigned to each analysis record
file_name	String	Original name of the uploaded image or video frame
input_type	String	Indicates whether the input is an image, batch image, camera image, or video frame
date_time	DateTime	Date and time when the analysis was performed
image_width	Integer	Width of the processed image in pixels
image_height	Integer	Height of the processed image in pixels
fruit_area	Integer	Number of pixels contained within the final banana mask
green_pixels	Integer	Number of segmented banana pixels classified as green
yellow_pixels	Integer	Number of segmented banana pixels classified as yellow
brown_pixels	Integer	Number of segmented banana pixels classified as brown
green_percentage	Float	Percentage of banana surface classified as green
yellow_percentage	Float	Percentage of banana surface classified as yellow
brown_percentage	Float	Percentage of banana surface classified as brown
ripeness_stage	String	Final ripeness classification: unripe, ripe, overripe or spoiled.
defect_count	Integer	Number of valid connected defect regions
defect_area	Integer	Total area of all detected surface defects in pixels

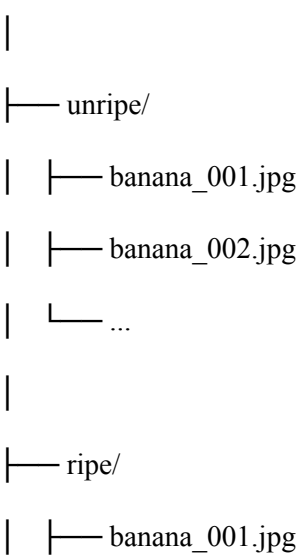
defect_percentage	Float	Percentage of the banana surface classified as defective
quality_grade	String	Final quality grade assigned by the rule-based grading module
shelf_life_days	Integer	Estimated number of remaining shelf-life days
recommendation	String	Suggested action, such as sale, quick consumption, discounting, or processing
processing_time	Float	Total execution time required to analyse the image
status	String	Indicates whether processing was successful or whether an error occurred

The data dictionary ensures that every inspection record follows a consistent structure. It also supports dashboard visualisation, historical comparison, batch processing summaries, and automated PDF report generation. The current project scope allows records to be stored in CSV format or in a lightweight SQLite database. The existing report identifies data management, dashboard analysis, batch processing, video processing, and PDF reporting as integrated parts of the proposed system.

3.2.5 Directory Organisation

The image collection should be organised into separate folders according to ripeness category. A proposed directory structure is shown below.

dataset/



```
│   ├── banana_002.jpg
│   └── ...
│
├── overripe/
│   ├── banana_001.jpg
│   ├── banana_002.jpg
│   └── ...
│
├── spoiled/
│   ├── banana_001.jpg
│   ├── banana_002.jpg
│   └── ...
│
└── self_collected/
    ├── controlled_background/
    ├── variable_lighting/
    ├── bruised_samples/
    └── video_frames/
```

This structure allows images to be selected according to their expected ripeness condition during system calibration and evaluation. It also simplifies the calculation of confusion matrices and classification accuracy during the result and discussion stage.

3.2.6 Data Preparation

Before an image is passed to the main processing pipeline, several preparation operations are performed. First, the file extension is checked to ensure that the uploaded file is supported. The image is then decoded and verified to confirm that it is not corrupted. Images with invalid dimensions or missing colour channels are rejected.

Accepted images are resized while preserving the aspect ratio. A maximum dimension such as 800 or 1,024 pixels may be used to reduce processing time without removing essential peel details. The

resized image is stored in BGR format because this is the default representation used by OpenCV. It is subsequently converted into other colour spaces depending on the processing stage.

A Gaussian filter is applied to reduce minor image noise. CLAHE is then applied to the luminance component of the image to improve local contrast without excessively altering peel colour information. After enhancement, the image is converted into HSV format for banana segmentation, ripeness analysis, and defect detection.

No artificial alteration of the final test images is performed. Operations such as rotation, brightness modification, or colour adjustment may be used during early parameter calibration, but the final evaluation should contain unmodified images to ensure that the reported performance reflects realistic system behaviour.

3.2.7 Ground-Truth Labelling

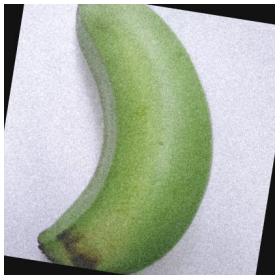
To evaluate the accuracy of the SmartBanana system, each selected test image should be assigned a ground-truth ripeness label. The ground-truth category may be determined through agreement between group members using the visible peel characteristics shown in Table 3.1. Images with uncertain or ambiguous ripeness conditions should be reviewed collectively before a final label is assigned.

Defect evaluation requires a separate ground-truth procedure. For a representative subset of images, visible defect regions may be manually outlined using an image annotation tool. The manually marked regions can then be compared with the automatically generated defect mask. This comparison enables the calculation of segmentation metrics such as Intersection over Union, precision, recall, and Dice coefficient.

The use of manually reviewed labels provides a reference against which the automated output can be evaluated. However, the labels remain based on visible external characteristics and should not be interpreted as laboratory measurements of internal fruit condition.

3.2.8 Sample Dataset Image

Table 3.3: Sample Dataset Image

Unripe	Ripe	Overripe	Spoiled
			

3.2.9 Dataset Limitations

The selected dataset and self-collected image collection contain several limitations. The images may differ in camera quality, resolution, background, illumination, banana variety, and viewing angle. Furthermore, the provided class labels may not follow an identical physiological ripeness scale across all image sources. External appearance may also be affected by bruising, shadows, surface reflections, and dirt, which can resemble natural brown ripening spots.

The system is therefore designed primarily for clearly visible single bananas captured under reasonably controlled conditions. Results obtained from heavily overlapping bananas, incomplete fruit images, extreme lighting, or complex backgrounds may be less reliable. These limitations are addressed through input guidance, contrast enhancement, morphological refinement, minimum-area filtering, and clear reporting of segmentation failures.

Overall, the combined public and self-collected image collection provides sufficient visual variation to calibrate and evaluate the SmartBanana image processing pipeline. The dataset supports the assessment of ripeness classification, banana segmentation, surface defect quantification, quality grading, processing efficiency, and robustness under different acquisition conditions. This methodology is aligned with the report's proposed classical processing approach and its focus on transparent, measurable image features.

3.3 Applications of the algorithm(s)

The SmartBanana system integrates a series of classical image processing algorithms to perform banana ripeness classification and quality assessment. Each algorithm is responsible for a specific stage of the processing pipeline, beginning with image acquisition and ending with quality recommendation and report generation. Unlike machine learning approaches, the proposed methodology uses deterministic algorithms that allow every intermediate processing result to be visualised and interpreted. Consequently, the final ripeness classification and quality grading can be directly explained using measurable image features such as colour distribution, defect area, and object geometry.

3.3.1 Image Acquisition

The first stage of the SmartBanana system acquires banana images from the user. Three acquisition methods are supported:

- Single image upload
- Batch image upload
- Video file processing

The system first validates the uploaded file format before decoding the image using OpenCV.

Supported formats include:

- JPG
- JPEG
- PNG

- MP4 (video)

Invalid or corrupted files are rejected before further processing.

The decoded image is stored as a three-channel BGR image because OpenCV internally represents colour images using the BGR colour space.

Sample Input: banana001.jpg
Sample Output:

- Loaded image
- Width = 1024 pixels
- Height = 768 pixels
- Channels = 3

Pseudocode:

```
Algorithm 1 Image Acquisition
Input: Image File
Output: BGR Image

Begin
        Read image
        If image cannot be loaded
        Display error
        Terminate
        Return image
End
```

3.3.2 Image Preprocessing

Before banana segmentation, the acquired image undergoes preprocessing to improve image quality and reduce unwanted variations.

Three preprocessing operations are performed:

1. Image resizing
2. Gaussian filtering
3. CLAHE enhancement

Image resizing standardises image dimensions while reducing computational cost.

Gaussian filtering removes small sensor noise without significantly affecting object boundaries.

The Gaussian filter is defined as

$$G(x,y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}}$$

where

- x, y represent pixel coordinates
- σ controls smoothing strength

A 5 × 5 Gaussian kernel is applied.

After noise removal, Contrast Limited Adaptive Histogram Equalisation (CLAHE) enhances local contrast while preventing excessive amplification of image noise.

Unlike standard Histogram Equalisation, CLAHE divides the image into small tiles before performing local histogram equalisation.

This improves visibility of

- bruises
- peel texture
- dark spots
- colour transitions

without significantly altering banana colour.

Sample Input: Original RGB image
Sample Output: Enhanced image with improved contrast

Pseudocode:

```
Algorithm 2 Image Preprocessing
Input: Original Image
Output: Enhanced Image

Resize image
Apply Gaussian Blur
Apply CLAHE
Return Enhanced Image
```

3.3.3 HSV Colour Conversion

Since RGB is highly affected by illumination, the enhanced image is converted into HSV colour space.

HSV separates

- Hue
- Saturation
- Value

allowing colour information to be analysed independently from brightness.

Hue values represent the actual peel colour.

Green bananas exhibit lower hue values.

Yellow bananas exhibit larger hue values.

Brown and black regions produce low saturation and low brightness.

OpenCV performs colour conversion using

```
cv2.cvtColor(image, cv2.COLOR_BGR2HSV)
```

Pseudocode:

Algorithm 3 HSV Conversion Input: Enhanced Image Convert RGB to HSV Return HSV Image

3.3.4 Banana Segmentation Using HSV Thresholding

Banana segmentation isolates the fruit from the surrounding background.

The HSV image is thresholded according to predefined lower and upper colour limits.

Each pixel satisfies

$$Lower \leq Pixel \leq Upper$$

Pixels inside the threshold become white

Pixels outside become black

The resulting binary image contains the banana mask.

HSV thresholding performs better than RGB thresholding because brightness variation has less influence on Hue.

Sample Output: Binary banana mask

Pseudocode:

```
Algorithm 4 HSV Thresholding
Input: HSV Image

For every pixel
        If HSV within threshold
                Mask = White
        Else
                Mask = Black
Return Binary Mask
```

3.3.5 Morphological Processing

Thresholding often introduces

- isolated pixels
- holes
- fragmented boundaries

Morphological Opening removes isolated foreground noise.

Opening consists of

$$Opening = Erosion \oplus Dilation$$

Morphological Closing repairs small holes.

$$Closing = Dilation \ominus Erosion$$

The structuring element is a 5 × 5 elliptical kernel.

These operations produce a cleaner banana mask before contour extraction.

Sample Output: Clean segmentation mask

Pseudocode:

```
Algorithm 5 Morphology
Input: Binary Mask

Apply Opening
```

```
Apply Closing
Return Clean Mask
```

3.3.6 Contour Detection and Banana Isolation

Contours are extracted from the refined binary mask.

Each contour represents one connected object.

Since only one banana is analysed at a time, the largest contour is selected.

The contour is used to calculate

- banana area
- banana boundary
- banana mask

Pixels outside the contour are discarded.

This ensures that background pixels do not influence ripeness analysis.

Pseudocode:

```
Algorithm 6 Contour Detection
Input: Clean Binary Mask

Detect Contours
Find Largest Contour
Generate Banana Mask
Return Banana
```

3.3.7 Ripeness Analysis

The segmented banana is analysed according to peel colour.

Three HSV masks are generated:

- Green
- Yellow
- Brown

The percentage of each colour is calculated by

$$Percentage = \frac{ColourPixels}{TotalBananaPixels} \times 100$$

The dominant colour determines the ripeness stage.

Example:

Green	Yellow	Brown	Result
80%	18%	2%	Unripe
15%	82%	3%	Ripe
5%	55%	40%	Overripe
2%	15%	83%	Spoiled

Pseudocode:

```
Algorithm 7 Ripeness Analysis
Input: Banana Mask

Calculate Green %
Calculate Yellow %
Calculate Brown %
Determine Dominant Colour
Return Ripeness Stage
```

3.3.8 Defect Detection

Dark regions are extracted from the segmented banana.

Binary thresholding identifies potential defects.

Connected Component Analysis labels every connected defect.

Small components below a minimum area threshold are removed.

The total defect area is

$$Defect\ Percentage = \frac{DefectArea}{BananaArea} \times 100$$

This percentage provides an objective measurement of banana quality.

Pseudocode:

```
Algorithm 8 Defect Detection
Input : Banana

Threshold Dark Regions
Morphology
Connected Components
Calculate Defect Area
Return Defect Percentage
```

3.3.9 Quality Grading and Shelf-Life Prediction

The final grading combines

- ripeness stage
- defect percentage

A rule-based decision engine assigns

- Grade A
- Grade B
- Grade C

The same information is used to estimate the remaining shelf life.

Example rules include:

- **Grade A:** Minimal defects with acceptable ripeness; suitable for normal sale.
- **Grade B:** Moderate defects or advanced ripeness; suitable for discounted sale or early consumption.
- **Grade C:** Severe defects or spoiled condition; recommended for processing or disposal if unsafe.

Shelf-life estimation follows the identified ripeness category, with unripe bananas having the longest remaining shelf life and spoiled bananas the shortest.

Pseudocode:

```
Algorithm 9 Quality Assessment
Input: Ripeness Stage, Defect Percentage

Assign Quality Grade
Estimate Shelf Life
Generate Recommendation
Return Results
```

3.3.10 Dashboard and Report Generation

After processing is complete, the SmartBanana system presents all analysis results through the dashboard. The dashboard displays the original image, enhanced image, segmentation mask, ripeness stage, defect percentage, quality grade, shelf-life estimation, and recommendations. Historical inspection records are stored to allow users to compare previous analyses over time.

The system also supports batch image processing, enabling multiple banana images to be analysed consecutively. For each processed image, a PDF report can be generated summarising all inspection results, including intermediate processing outputs and the final assessment. This reporting functionality provides users with a permanent record of the inspection process and supports quality monitoring in commercial or educational environments.

Pseudocode:

```
Algorithm 10 Report Generation
Input: Analysis Results

Display Dashboard
Store Record
Generate PDF Report
Return Report
```

Result & Discussion

4.1 Experimental Results

4.1.1 Evaluation design and test data

At the time of testing, the repository contained 11,793 images in four folders: unripe, ripe, overripe and spoiled. A filename audit found that some folders included generic or contradictory source names, especially the spoiled folder, which also contained images whose names referred to ripe or unripe bananas. To reduce label noise, the quantitative evaluation used only source groups whose filenames explicitly agreed with the intended class. Roboflow augmentation suffixes were removed to identify the original source group. A deterministic seed of 2133 selected 100 source groups per class, and only one augmented representative from each group was analysed. The final evaluation set therefore contained 400 images with equal class support.

Table 4.1 Repository composition and source-grouped evaluation sample

Class	Repository images	Clean source groups	Evaluation images
Unripe	1,902	634	100
Ripe	3,522	793	100
Overripe	2,349	783	100
Spoiled	4,020	613	100
Total	11,793	2,823	400

Each image passed through the same automatic pipeline described in Chapter 3: resizing, Gaussian smoothing, CLAHE contrast enhancement, HSV conversion, colour thresholding, morphological opening and closing, largest-contour selection, peel-colour measurement, connected-component defect measurement, and rule-based stage assignment. The manual boundary was disabled during the 400-image comparison so that every sample received the same automatic treatment.

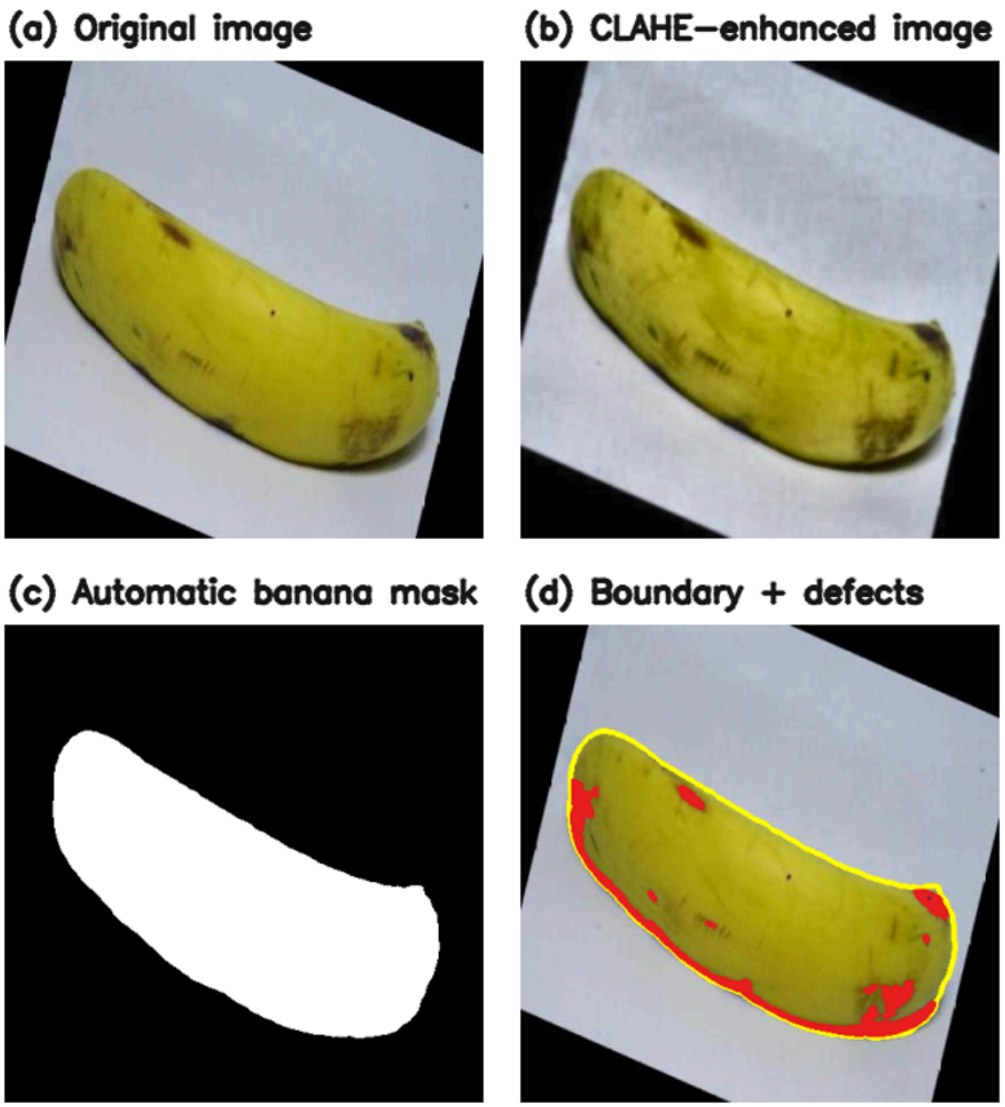


Figure 4.1 Representative stages of the automatic processing pipeline for a correctly classified ripe banana.

4.1.2 Classification performance

Performance was calculated from the confusion matrix using accuracy = number of correct predictions / total predictions. For each class, precision measures the proportion of predictions for that class that were correct, recall measures the proportion of actual samples of that class that were recovered, and F1-score is the harmonic mean of precision and recall. All 400 images produced a valid segmentation and result; no evaluation image failed processing.

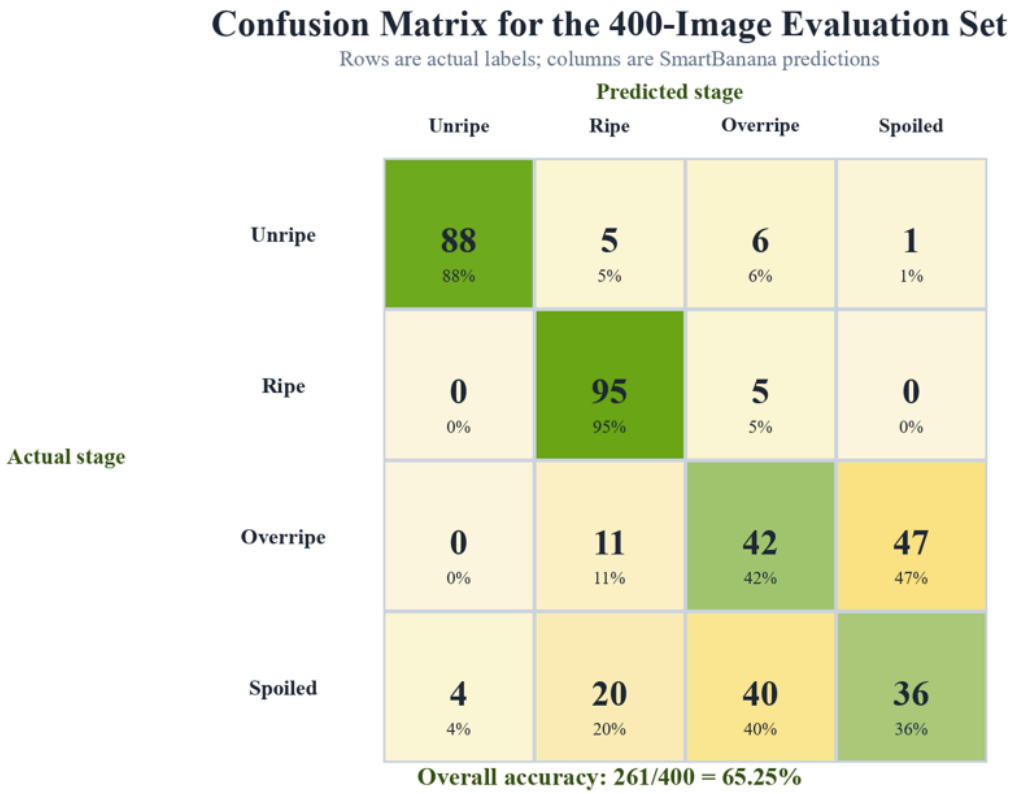


Figure 4.2 Confusion matrix for the calibrated rule set. Each row contains 100 actual images.

The system classified 261 of 400 images correctly, giving an overall accuracy of 65.25%. Unripe and ripe bananas produced the strongest recalls, at 88% and 95% respectively. The late-ripeness classes were more difficult: overripe recall was 42% and spoiled recall was 36%. The most frequent late-stage errors were 47 overripe images predicted as spoiled and 40 spoiled images predicted as overripe.

Table 4.2 Class-wise classification metrics

Class	Precision ↑ (%)	Recall ↑(%)	F1-score ↑ (%)	Support
Unripe	95.65	88.00	91.67	100
Ripe	72.52	95.00	82.25	100
Overripe	45.16	42.00	43.52	100
Spoiled	42.86	36.00	39.13	100
Macro average	64.05	65.25	64.14	400

Table 4.2 presents the precision, recall and F1-score obtained for each ripeness stage using 100 test images per class. The upward arrows indicate that higher values represent better classification performance. Precision measures the reliability of predictions for a class, recall measures the proportion of actual samples detected, and the F1-score balances both measures.

The **Unripe** class achieved the strongest overall performance, with 95.65% precision, 88.00% recall and a 91.67% F1-score. This means that 88 of the 100 unripe bananas were correctly detected, while predictions labelled as unripe were rarely incorrect. The distinctive green peel colour provides strong separation from the other ripeness stages.

The **Ripe** class recorded the highest recall at 95.00%, showing that 95 of the 100 ripe bananas were correctly identified. However, its lower precision of 72.52% indicates that some overripe or spoiled bananas were incorrectly classified as ripe. Its F1-score of 82.25% therefore remains high but is lower than that of the unripe class.

Performance decreased considerably for the **Overripe** class. Its precision, recall and F1-score were 45.16%, 42.00% and 43.52%, respectively. Only 42 of the 100 overripe samples were correctly detected. This suggests substantial confusion with neighbouring stages, particularly ripe and spoiled, because ripeness develops gradually and their yellow, brown and dark regions can overlap.

The **Spoiled** class produced the weakest results, with 42.86% precision, 36.00% recall and a 39.13% F1-score. Only 36 of the 100 spoiled bananas were correctly identified. Variations in browning, mould, illumination and defect coverage may cause spoiled bananas to resemble overripe samples. The rule-based colour thresholds may therefore be insufficient to represent the wide visual variation of spoilage.

The macro-average precision, recall and F1-score were 64.05%, 65.25% and 64.14%, respectively. Because all four classes contain an equal number of test images, the macro recall of 65.25% also corresponds to **261 correctly classified images out of 400**. These results indicate moderate overall performance but substantial differences between classes.

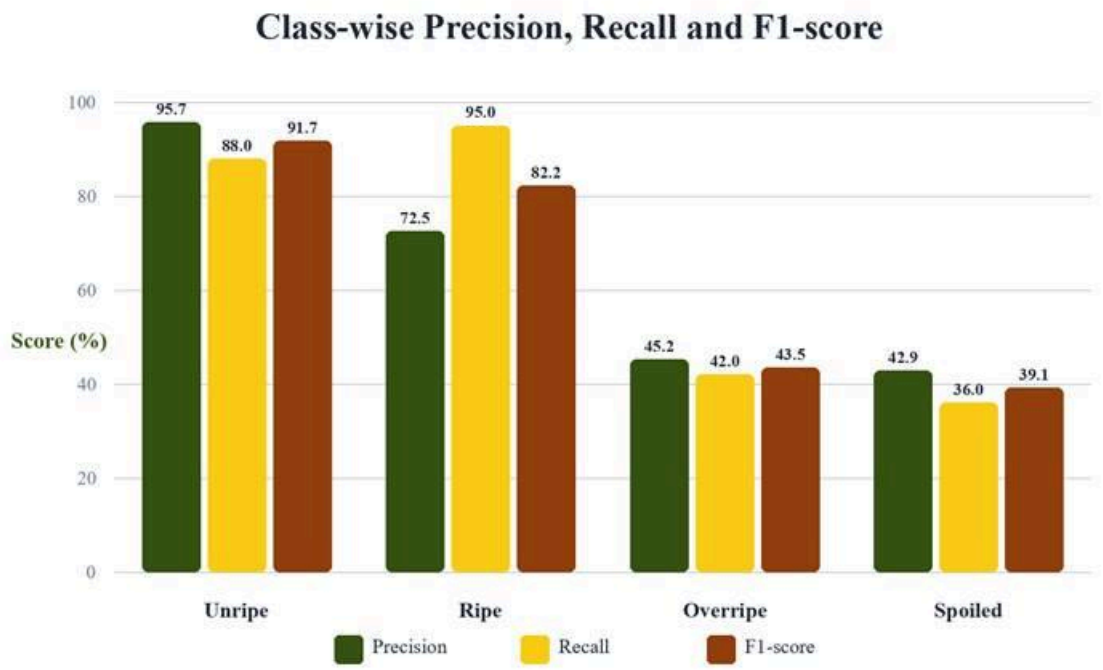


Figure 4.3 Precision, recall and F1-score by ripeness stage.

Figure 4.3 visually reinforces this interpretation. The bars show strong performance for unripe and ripe bananas, followed by a sharp reduction for the overripe and spoiled stages. Consequently, the system is more dependable for identifying early ripeness stages than for distinguishing advanced deterioration. Future improvements should focus on separating overripe and spoiled bananas through better threshold calibration, additional texture and defect features, more varied training or calibration images, and improved control of lighting and background conditions.

4.1.3 Effect of rule calibration

The initial prototype used a green threshold of 40% for unripe fruit, identified spoiled fruit when defect area reached 50% or brown area reached 30%, and otherwise separated overripe fruit at 25% defect or 20% brown area. Inspection of source-grouped feature distributions showed that this arrangement marked many naturally speckled overripe bananas as spoiled and missed partially green unripe bananas. The calibrated rules lowered the unripe green threshold to 20%, required stronger evidence for spoiled fruit, added an explicit ripe condition of at least 65% yellow with less than 25% defect area, and retained an intermediate overripe decision.

Table 4.3 Accuracy comparison between the initial and calibrated rule sets

Rule set	Correct / 400	Accuracy ↑ (%)	Change
Initial prototype	249 / 400	62.25	Reference
Calibrated	261 / 400	65.25	+3.00 percentage points

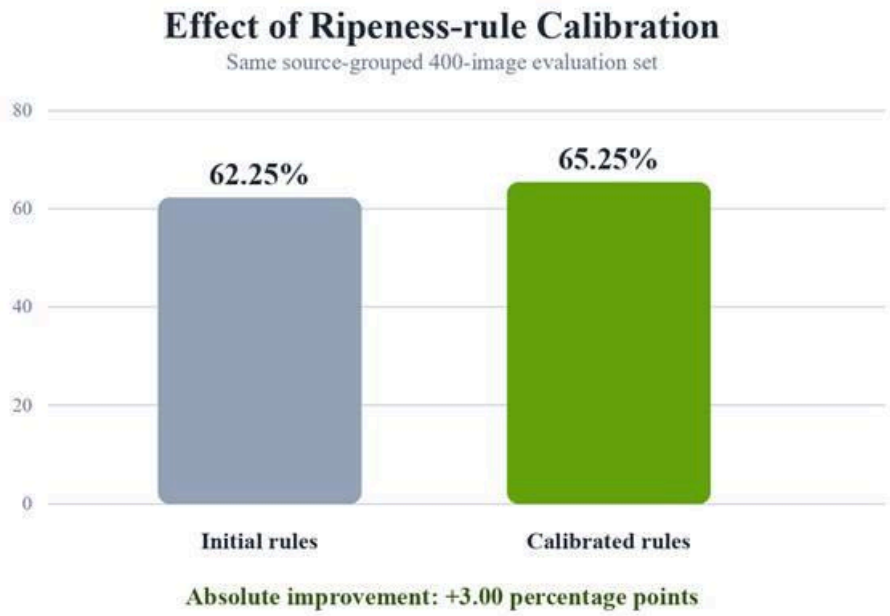


Figure 4.4 Overall accuracy before and after rule calibration on the same evaluation set.

4.1.4 Colour and defect characteristics

The median feature values in Table 4.4 explain why the first two classes were easier to recognise. Unripe bananas contained a median of 84.0% green pixels, whereas ripe bananas contained a median

of 89.6% yellow pixels. In contrast, the overripe and spoiled groups occupied overlapping regions: both contained almost no green, substantial yellow, brown peel and large dark-defect areas. Spoiled images were not consistently darker than overripe images; the median defect area was 36.9% for spoiled fruit compared with 46.1% for overripe fruit in this sample. This overlap places a practical limit on a fixed colour-threshold classifier.

Table 4.4 Median peel-colour and defect percentages in the evaluation set

Actual class	Green (%)	Yellow (%)	Brown (%)	Defect (%)
Unripe	84.0	10.6	2.4	29.1
Ripe	0.1	89.6	4.4	13.1
Overripe	0.1	55.7	33.1	46.1
Spoiled	0.1	66.7	26.6	36.9

4.1.5 Processing efficiency and functional verification

The mean end-to-end processing time within the analysis function was 0.0175 s per image, the median was 0.0169 s, and the 95th percentile was 0.0224 s. The recorded range was 0.0092–0.1789 s. These measurements cover resizing, enhancement, segmentation, feature extraction, defect analysis and classification; they exclude browser upload and page-rendering time. The result indicates that the classical pipeline is suitable for interactive single-image use and sampled-frame video analysis on the test computer.

Table 4.5 Functional and regression verification of the completed system

Test area	Verification performed	Result
Automatic image analysis	Valid JPG/PNG is segmented and assigned ripeness, grade, shelf life and recommendation	Pass
Input validation	Empty, invalid or unusable images produce controlled messages instead of application crashes	Pass
Manual boundary	Mouse-drawn closed region is converted to a binary mask and re-runs the same analysis	Pass
Segmentation correction	Synthetic foreground distractor test confirms that a user boundary can replace an incorrect largest contour	Pass
Batch/video/history	Batch records, sampled video frames, SQLite history and CSV/PDF export paths execute	Pass
Automated regression	Four unit tests completed successfully	4/4 pass

Streamlit smoke test	All application pages load without reported exceptions; live health endpoint returns HTTP 200	Pass

The optional manual-boundary function is presented after automatic masking. The user draws a closed banana region with the mouse; the RGBA drawing layer is converted to a binary mask, resized to the analysis dimensions, and reduced to its largest valid contour. The colour and defect features are then recomputed inside that user-defined region. This design preserves the speed of automatic segmentation for normal cases while giving the user direct control when a similarly coloured background object or connected shadow distorts the mask.

4.2 Discussion and Critical Interpretation

4.2.1 Interpretation of the main findings

The results demonstrate that HSV colour proportions are effective for the visually distinct ends of the early ripening sequence. Unripe bananas are dominated by green hue, while ripe bananas are dominated by yellow hue; this produced high recall without a trained model. Gaussian smoothing reduces isolated sensor noise, CLAHE improves local contrast under uneven illumination, and morphological opening and closing remove small false regions and fill gaps. Largest-contour isolation further simplifies the scene by treating the principal banana-coloured object as the fruit. Together, these methods provide fast and explainable predictions: every decision can be traced to measured green, yellow, brown and defect percentages.

The 3.00-percentage-point improvement over the initial thresholds confirms that calibration matters, particularly for partially green fruit and the boundary between normal speckling and serious deterioration. However, the modest size of the improvement also shows that threshold tuning alone cannot resolve the late-stage classes. Overripe and spoiled bananas form a visual continuum rather than two perfectly separated colour clusters. Brown patches, shadows, bruises and mould can all contribute dark pixels, while a spoiled banana may still contain a large yellow region. Consequently, the connected-component defect measure is useful as evidence of damage but is not a direct measurement of edibility or microbiological safety.

4.2.2 Accuracy limitations and sources of error

The principal classification limitation is label and feature overlap. The repository folders are not fully consistent with their filenames, and some generic banana filenames provide no independent confirmation of ripeness. Restricting evaluation to explicit source names reduces obvious contradictions, but filename-derived labels remain weaker than expert inspection. In addition, augmented photographs share the same original content; source grouping prevents duplicate counting in the 400-image set, yet the evaluation is still internal to one repository and should not be interpreted as external-field accuracy.

Segmentation can also affect every downstream percentage. The automatic HSV range intentionally includes green, yellow, orange and brown peel, but similarly coloured objects may enter the mask. The largest-contour rule can select the wrong object when the background region is larger than the banana, and very dark spoiled peel may fall below the saturation or value limits. Illumination, camera white balance, reflections and deep shadows shift HSV values even after CLAHE. Because no pixel-level ground-truth masks were supplied, segmentation quality could not be quantified using Intersection over Union or Dice score. The manual boundary reduces this failure mode but does not solve genuine overlap between overripe and spoiled peel.

The displayed confidence value is a heuristic based on separation between the dominant colour percentages and their combined coverage. It is therefore an indication of decision clarity, not a calibrated probability. Similarly, shelf-life days and recommendations are operational rules derived from ripeness and quality grade; they should be treated as guidance rather than laboratory predictions. A user should still reject fruit showing mould, leakage, unusual odour or other safety concerns even when the image classification is less severe.

4.2.3 Practical implications of the completed system

For the assignment objectives, the completed system successfully demonstrates an end-to-end, explainable inspection workflow. It accepts images, camera captures, ZIP batches and videos; segments the banana; quantifies peel colours and visible defects; assigns a ripeness stage and quality grade; estimates shelf life; stores inspection history; and exports CSV or PDF evidence. The optional mouse-drawn boundary is especially valuable in a teaching or small-retail context because the operator can see and correct the region used by the algorithm instead of receiving an unexplained result.

The current system is most reliable as a screening and decision-support tool for distinguishing unripe and ripe fruit. Predictions of overripe versus spoiled should be interpreted with greater caution and reviewed alongside the segmentation overlay and defect percentage. This division of responsibility is appropriate for a transparent prototype: automation handles repetitive measurement, while a human remains responsible for ambiguous late-stage and food-safety decisions.

4.2.4 Recommended improvements

Further improvement should begin with data quality rather than additional threshold complexity. A new test set should use expert-verified stage labels, controlled capture metadata and manual banana masks. This would allow separate reporting of segmentation IoU/Dice and classification accuracy. Colour calibration using a reference card or grey-world correction could reduce camera-dependent hue shifts. Texture descriptors, local colour histograms and the spatial distribution of dark spots could help distinguish normal overripe speckling from extensive decay.

If a learning-based extension is permitted, a compact transfer-learning classifier or segmented-feature model could be trained with source-grouped train, validation and external test splits. The manual boundaries collected during normal use could become reviewed mask annotations for later segmentation improvement. Any learned model should retain visual overlays and per-class probabilities so that the interpretability achieved by the classical pipeline is not lost.

4.2.5 Chapter summary

In summary, SmartBanana achieved 65.25% accuracy on a balanced, source-grouped 400-image internal evaluation and processed an image in 0.0169 s at the median. The system was strongest for unripe and ripe fruit and less reliable for the overlapping overripe and spoiled stages. Calibration improved overall accuracy by 3.00 percentage points, while the optional manual boundary provides a practical correction for segmentation errors. These results support the usefulness of the implemented classical image-processing workflow while clearly identifying the data, segmentation and late-stage classification limitations that should guide future work.

Conclusion

5.1 Achievements

SmartBanana was completed as an end-to-end, classical image-processing decision-support system. The final application accepts image uploads and camera captures, supports optional mouse-drawn banana boundaries, processes ZIP batches and sampled video frames, measures peel colour and visible defects, assigns ripeness and quality results, generates handling recommendations, stores inspection history, and exports CSV and PDF evidence. In the balanced source-grouped evaluation, the system processed all 400 images without a segmentation failure and correctly classified 261 images, producing 65.25% overall accuracy. Median analysis time was 0.0169 s per image, while all four automated regression tests and the Streamlit smoke tests passed.

Table 5.1 Achievement status of the objectives stated in Section 1.2

Objective	Status	Experimental and implementation evidence
1	Partially Fulfilled	Automatic HSV-based ripeness assessment was completed and processed all 400 evaluation images. Overall accuracy was 65.25%; recall was 88% for unripe and 95% for ripe, but only 42% for overripe and 36% for spoiled.
2	Fulfilled	The system detects dark surface regions, calculates defect percentage, assigns quality grades, and displays a defect overlay. Every evaluation image produced a valid result, and the related regression tests passed.
3	Fulfilled	Shelf-life guidance, recommendations, dashboard visualisations, history, batch and video processing, CSV export, and automated PDF reports were implemented and passed functional smoke testing.

Project Code Repository:
https://github.com/j1ngy0ng/Image-Processing---Banana-Ripeness-AI-Detection.git

5.1.1 Objective 1: Automatic ripeness assessment

Status: Partially Fulfilled. The required automatic ripeness-assessment workflow was successfully developed using HSV colour analysis. The completed pipeline resizes and enhances the image, isolates the largest banana-coloured contour, calculates green, yellow and brown peel percentages, and assigns one of four stages: unripe, ripe, overripe or spoiled. Experimental results confirm that the system is capable of automatic stage determination, achieving 65.25% overall accuracy and no processing failures on the 400-image sample.

The objective is assessed as partially rather than fully fulfilled because performance was not equally reliable across all stages. Unripe and ripe recall reached 88% and 95%, but overripe and spoiled recall reached only 42% and 36%. The optional manual boundary corrects segmentation errors caused by distracting backgrounds, but it cannot remove the genuine colour and defect overlap between late-stage fruit. The system therefore satisfies the implementation aspect of the objective and performs

well for early stages, while further classification improvement is required before consistent four-stage deployment can be claimed.

5.1.2 Objective 2: Defect detection and quality grading

Status: Fulfilled. SmartBanana identifies dark and brown surface regions inside the segmented banana, removes small noise through morphology, groups meaningful regions using connected-component analysis, and reports both defect count and defect-area percentage. The same measurements are shown in the red defect overlay and converted into quality grades A, B or C through transparent rules. This provides the defect quantification and rule-based grading specified by Objective 2 for every valid analysis.

The achievement is functional and explainable: the user can inspect the boundary, overlay and percentages behind the assigned grade. Nevertheless, the grade represents visible external appearance rather than laboratory confirmation of internal damage or food safety. This boundary is treated as a limitation of interpretation, not as a missing function in the stated objective.

5.1.3 Objective 3: Decision support, visualisation and reporting

Status: Fulfilled. The completed application translates ripeness and defect measurements into estimated shelf-life days and practical handling recommendations. It presents analysis summaries and processing evidence, maintains SQLite inspection history, supports dashboard review, and creates downloadable CSV and PDF reports. Batch-image and sampled-video workflows extend the same outputs to multiple records. These implemented and smoke-tested functions directly satisfy the four output categories specified in Objective 3: shelf-life estimation, recommendations, analytical visualisations and automated reporting.

5.1.4 Best-performing algorithm configuration

Among the tested configurations, the calibrated HSV-based classical pipeline produced the best experimental result. The initial rule set achieved 62.25% accuracy, whereas the calibrated decision rules achieved 65.25% on the same 400-image evaluation set, an absolute improvement of 3.00 percentage points. The best configuration combines Gaussian smoothing, CLAHE enhancement, HSV thresholding, morphological opening and closing, largest-contour isolation, colour-percentage analysis, connected-component defect measurement, and the calibrated ripeness rules.

This conclusion concerns the rule configurations implemented and compared in this project. It does not claim that the classical method is universally more accurate than convolutional neural networks or other learning-based approaches. Its main advantages for the present prototype are low processing cost, rapid response, deterministic behaviour, and the ability to explain each decision using visible masks and measured percentages.

5.1.5 Project code repository

The completed source code, application files and project dataset structure are available at https://github.com/j1ngy0ng/Image-Processing---Banana-Ripeness-AI-Detection.

5.2 Limitations and Future Works

5.2.1 Current limitations

Late-stage class overlap. Overripe and spoiled bananas share yellow peel, brown patches and extensive dark regions. In the evaluation, 47 overripe images were predicted as spoiled and 40 spoiled images were predicted as overripe. Fixed thresholds cannot fully represent this continuous biological transition, and external appearance alone cannot confirm mould, odour, leakage or internal tissue condition.

Segmentation and illumination sensitivity. The automatic HSV mask may include similarly coloured background objects, lose very dark peel, or select the wrong largest contour. CLAHE improves local contrast but does not completely standardise camera white balance, exposure, reflections and shadows. The manual boundary provides a correction path, but it requires operator effort and is not available as ground-truth segmentation for the entire dataset. Consequently, IoU and Dice scores could not be reported.

Heuristic output rules. Quality grade, confidence, shelf-life days and recommendations are deterministic operational rules. The confidence value is not a calibrated probability, and the shelf-life estimate does not account for cultivar, temperature, humidity, storage gas, handling damage or the time already spent after harvest. These outputs should support, rather than replace, human judgement and food-safety inspection.

Prototype deployment boundaries. Video analysis samples frames independently instead of tracking individual fruit, the SQLite history is intended for local use, and performance has not been validated in a multi-user production environment. Browser upload, camera permissions and reporting speed also depend on the user's device and network context, even though the core image-analysis function is fast.

5.2.2 Future works

Create an expert-verified benchmark. A new dataset should record cultivar, capture device, illumination, storage condition and expert-confirmed ripeness. Original photographs must be grouped before creating development, validation and external test splits. Pixel-level banana masks and defect annotations would allow segmentation IoU, Dice score and defect-localisation accuracy to be reported separately from classification accuracy.

Improve capture normalisation and segmentation. A colour reference card, grey-world correction or controlled imaging enclosure could reduce hue variation. The automatic mask should be compared with GrabCut, watershed segmentation and a compact learned segmentation model. Reviewed manual boundaries collected through the current interface could form the initial mask-annotation set for this comparison.

Strengthen late-stage classification. Future experiments should add texture descriptors, local colour histograms, spot size and spatial-distribution features to the current global colour percentages. A lightweight transfer-learning classifier may then be compared against the calibrated classical baseline using the same source-grouped splits. An uncertainty or human-review outcome should be introduced when overripe and spoiled scores are too close for a defensible automatic decision.

Calibrate quality and shelf-life outputs. Quality-grade thresholds should be reviewed with domain experts, while confidence should be calibrated against validation outcomes. Shelf-life prediction

should incorporate storage temperature, humidity, cultivar and repeated observations over time. A longitudinal experiment that photographs the same bananas daily would provide stronger evidence than a single external image.

Prepare for practical deployment. A future version should add tracked video objects, configurable user roles, central database storage, audit logs and structured user-acceptance testing with retailers or laboratory staff. Performance should be measured on low-cost devices and under realistic batch loads before the prototype is considered for operational inventory decisions.

5.2.3 Final conclusion

SmartBanana demonstrates that a transparent classical image-processing pipeline can integrate banana segmentation, ripeness assessment, visible-defect measurement, quality grading, decision support and reporting within one accessible application. The calibrated method achieved 65.25% accuracy and particularly strong recognition of unripe and ripe fruit, while its weaker separation of overripe and spoiled samples identifies a clear research priority. Overall, the project fulfilled the defect-analysis and decision-support objectives and partially fulfilled the four-stage ripeness objective. The completed prototype is therefore suitable as an explainable educational and screening system, with expert-labelled data, improved late-stage features and external validation required for higher-stakes deployment.

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Instruction to Student: Compile an exhaustive bibliography of all scholarly material referenced within this assignment. You must strictly adhere to the APA referencing format, ensuring each entry includes the author, publication year, title, and publisher or URL. Every in-text citation must have a matching full entry.

- *Specify the Digital Object Identifier (DOI) for your dataset where available.*
- *Avoid websites/forums/researchgate/arxiv if possible.*
- *Numbered lists are not permitted in this section.*
- *Organise all bibliographic entries in alphabetical order.*
- *Employ academic search engines or university library databases to ensure the precision and standardisation of your reference metadata.*

Appendix A

TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY
Group Contract Form

Faculty : Faculty of Computing and Information Technology (FOCS)
Programme :
Year of Study :
Semester :
Course Title : Image Processing
Course Code : BMDS2133
Group Assignment / Project Title :
Name of Tutor :
Tutorial Group :

Student Name and ID	Role(s)/Task(s) Assigned	Signature and Date

Appendix B

AI Usage Disclosure Form

Student Name : _____

Programme : _____

Course : _____

Assessment Title : _____

1. AI Tool(s) Used (Check that all that apply)

- ☐ None (I did not use any AI tools for this assignment)
- ☑ ~~ChatGPT (Version: _____)~~
- ☐ Deepseek
- ☐ Gemini
- ☐ Other: _____

2. Nature of Assistance (Check all that apply)

- ☐ **Brainstorming:** Generating ideas, topics, or outlines.
- ☑ ~~**Research:** Summarising long articles or finding concepts.~~
- ☐ **Editing:** Checking grammar, spelling, or sentence structure.
- ☑ ~~**Coding/Math:** Debugging code or explaining a formula.~~
- ☐ **Creation:** Generating images, data, or draft text.

3. Disclosure Statement

Provide a brief explanation of how AI tools were utilised in completing the task and describe the measures taken to ensure that the final submission reflects your own independent work.

Example:
"I used ChatGPT to structure my initial ideas into an outline. Subsequently, I composed the full paragraphs independently and cross-checked the referenced dates against my textbook to ensure accuracy."

Your Statement:

4. Verification & Integrity

- ☑ ~~I have reviewed and verified the accuracy of all facts, data, and citations generated with the assistance of AI tools.~~
- ☑ ~~I have properly acknowledged and cited any AI-generated text, ideas, or content in accordance with my instructor's requirements.~~
- ☑ ~~I confirm that the reasoning, interpretation, and analysis presented in the final submission are entirely my own.~~

Student Signature : _____

Date : _____

AMM 5.3.2026/SMC 1.4.2026